Türkiye at the Crossroads: Climate Change Impacts and Internal Migration Determinants

Sedat Ors*

March 2026

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Abstract

Climate change is increasingly recognized as an important driver of human mobility. This study examines how climatic conditions, economic factors, and demographic pressures shape internal migration across Türkiye using province-level migration flows between 2008 and 2024. Building on the Random Utility Model and the gravity framework, we analyze migration flows between provinces using three complementary econometric approaches: a three-way fixed effects specification, the Poisson Pseudo-Maximum Likelihood estimator, and the Hausman–Taylor estimator.

The results provide consistent evidence that climatic stress plays a significant role in shaping internal migration patterns. In particular, we find nonlinear temperature effects, indicating that migration responses intensify as temperatures deviate from moderate levels. Graphical and bin-based analyses reveal a thermal threshold: migration increases sharply once summer maximum temperatures exceed approximately 32°C. In addition, lower precipitation levels are associated with increased out-migration, particularly in provinces experiencing drought. Environmental shocks such as wildfire exposure further contribute to migration pressures, while electricity consumption appears to reflect both economic capacity and local adaptation mechanisms.

The analysis also highlights the role of demographic pressures. Refugee presence is associated with higher internal migration in several specifications, suggesting that refugee inflows may operate as a secondary exposure that amplifies mobility pressures in environmentally vulnerable regions. Subsample analyses further show that climate-related migration responses are strongest in provinces experiencing severe climatic stress.

Overall, the findings underscore the complex interaction between climatic variability, environmental shocks, economic conditions, and demographic pressures in shaping internal migration in Türkiye. These results highlight the importance of integrating climate risks and demographic dynamics when evaluating migration patterns and designing adaptation strategies in climate-vulnerable regions.

*PhD Candidate, School of Economics, Georgia Institute of Technology, Atlanta, GA, USA. Email: sors3@gatech.edu, ORCID: 0009-0006-1953-1604.

1 Introduction

Climate change is increasingly recognized as an important driver of human mobility, reshaping how researchers and policymakers understand population movements in response to environmental stress. A growing body of evidence shows that climate change contributes to rising temperatures (Roberts et al., 2013; Bohra-Mishra et al., 2014; Cattaneo and Peri, 2016; Cai et al., 2016; Hirvonen, 2016; Nawrotzki et al., 2017; Burzyński et al., 2022), changing precipitation patterns (Fisher, 1925; Munshi, 2003; Coniglio and Pesce, 2015; Cattaneo et al., 2019), and rising sea levels (Gray, 2009). These environmental changes are expected to increase the frequency and severity of natural disasters such as droughts, floods, wildfires, and storms (Findley, 1994; Halliday, 2006; Reuveny and Moore, 2009; Koubi et al., 2016; Thiede et al., 2016; Dallmann and Millock, 2017; Ruiz et al., 2017; Murray-Tortarolo and Salgado, 2021; IPCC, 2022). As climatic shocks alter agricultural productivity, livelihoods, and living conditions, affected populations may respond by migrating temporarily or permanently. However, the relationship between climate change and migration remains complex, reflecting the diverse economic, social, and environmental mechanisms through which climate variability influences human mobility (Tol, 2002; Laczko et al., 2009; Rummukainen, 2012; Joseph and Wodon, 2013; Noy, 2017). Understanding these dynamics is therefore critical for evaluating the socioeconomic consequences of climate change and designing effective adaptation strategies. As Cifci and Oliver (2018) notes, addressing climate change involves balancing urgent adaptation needs with non-trivial economic costs and potential impacts on development.

A large body of literature examines the relationship between environmental change and migration. Early work by El-Hinnawi (1985) introduced the concept of environmental migrants, referring to individuals who move due to sudden or gradual environmental changes.[1] Subsequent research emphasizes that migration responses to environmental shocks depend on complex interactions between environmental conditions, economic opportunities, demographic pressures, and political institutions (Myers, 1993; Sasser, 2010; Black et al., 2011; Dun, 2011; Özden et al., 2011; Piguet et al., 2011; Jennings and Gray, 2015; Gray et al., 2020; Cattaneo et al., 2019; Erdoğan and Cantürk, 2022; Thorn et al., 2023). These dynamics have generated increasing discussion of "climate refugees," highlighting how environmental shocks may produce large-scale displacement (Black, 2001; Brown, 2008; Saldaña-Zorrilla and Sandberg, 2009; Feng et al., 2010; Nawrotzki et al., 2015; Noy, 2017; Kaczan and Orgill-Meyer, 2020; Helbling and Meierrieks, 2021; IPCC, 2022). Estimates of future climate-induced migration vary widely. Myers (2002) warned that up to 200 million people could be displaced by 2050, while more recent projections suggest that over 143 million individuals in Sub-Saharan Africa, South Asia, and Latin America may migrate internally as a result of climate change (Rigaud et al., 2018). These projections highlight the growing importance of understanding migration as a potential adaptation mechanism to environmental shocks (Stern, 2006; Coniglio and Pesce, 2015; Cattaneo et al., 2019; Beine and Jeusette, 2021; Burzyński et al., 2022).

Climate change may also influence migration indirectly through its effects on economic conditions and conflict. A growing body of evidence shows that climatic shocks reduce agricultural productivity and increase poverty and economic vulnerability (Schlenker and Roberts, 2009; Cai et al., 2016; Cattaneo and Peri, 2016). Extreme temperatures and prolonged droughts can intensify these pressures and contribute to social instability and displacement (Kelley et al., 2017; Selby et al., 2017; Chen and Mueller, 2019; Erdoğan and Cantürk, 2022). A prominent example is the severe drought that affected Syria between 2007 and 2010. Kelley et al. (2015) show that this prolonged drought contributed to agricultural collapse, rural displacement, and economic hardship, factors that helped intensify the Syrian civil conflict and ultimately generated large refugee flows.

Türkiye has been directly affected by these regional dynamics. The country currently hosts nearly five million refugees, including approximately 3.5 million Syrians, according to the Republic of Türkiye's Ministry of Interior Presidency of Migration Management (RTMIPMM).[2] The arrival of such a large refugee population has increased pressure on housing markets, public services, and local labor markets in several regions of the country. These demographic pressures may influence the mobility decisions of the local population, generating additional internal migration flows (Erdoğan and Cantürk, 2022). Migration flows may also reshape social and cultural interactions between host and migrant populations through compositional change

[1]Environmental migrants are individuals or groups who, due to environmental changes that adversely affect their living conditions, are forced to leave their habitual residence either temporarily or permanently.

[2]Official statistics available at: https://en.goc.gov.tr/

and cultural diffusion (Rapoport et al., 2020). These mechanisms illustrate how climate-related shocks may generate secondary displacement effects, where the consequences of climate-induced migration in one region create additional migration pressures elsewhere (Barrios et al., 2006; Joseph and Wodon, 2013).

Against this background, this study examines how climatic conditions and refugee-related demographic pressures influence internal migration patterns in Türkiye between 2008 and 2024.[3] We focus on internal mobility because Turkish citizens face significant visa restrictions when attempting to migrate internationally, which function as institutional "border frictions" in gravity-type migration models (McCallum, 1995; Anderson and Van Wincoop, 2003; Silva and Tenreyro, 2006a; Anderson, 2013; Chen and Mueller, 2019). Using province-level migration flows between origin and destination cities, we analyze how climatic conditions, economic factors, and geographic frictions jointly shape migration decisions.

To capture heterogeneity in climatic exposure, the empirical analysis is conducted using three complementary samples. First, we examine the full national migration system covering all 81 provinces of Türkiye. Second, we analyze a subsample of 30 provinces that are more exposed to climatic variability and environmental stress.[4] Third, we focus on a narrower group of 17 drought-affected provinces identified using the Bagnouls–Gaussen aridity index developed by Cebeci et al. (2019).[5] This multi-sample design allows us to evaluate whether the relationship between climate conditions and migration becomes stronger in regions that are more vulnerable to climatic shocks.

Our empirical framework combines the Random Utility Model (RUM) with a gravity-type migration specification (Backhaus et al., 2015; Beine and Parsons, 2017). The gravity framework captures how temperature, precipitation, GDP levels, and geographic distance in both origin and destination cities shape migration flows, while the RUM provides the behavioral foundation for destination choice based on expected utility differences. The model is estimated using a three-way fixed effects specification, the Poisson Pseudo-Maximum Likelihood (PPML) estimator, and the Hausman–Taylor estimator to address zero migration flows, potential endogeneity, and unobserved heterogeneity (Silva and Tenreyro, 2006a; Hausman and Taylor, 1981; Wooldridge, 2010).

The results consistently demonstrate the importance of climatic, economic, and demographic factors in shaping internal migration in Türkiye. Higher temperatures in origin cities significantly increase outmigration, with particularly strong effects in drought-prone regions where extreme summer temperatures exceed approximately 32°C, indicating a nonlinear relationship between climatic stress and migration responses. Precipitation also influences migration dynamics, reflecting its role in shaping agricultural productivity and household income (Hirvonen, 2016). Climatic conditions in destination cities further influence migration decisions, as individuals tend to relocate to areas with milder temperatures and more favorable environmental conditions. Economic factors also play an important role: higher GDP levels in origin provinces are associated with greater migration flows, suggesting that economic capacity facilitates mobility, while higher GDP in destination provinces attracts migrants seeking better opportunities. Geographic distance remains a strong constraint, as longer distances significantly reduce migration flows due to higher migration costs (Beine and Parsons, 2015). Environmental shocks such as wildfires also contribute to migration pressures, while electricity consumption appears to capture both economic activity and adaptation capacity. Importantly, the magnitude of climate-related migration effects increases when the analysis focuses on provinces more exposed to climatic stress, particularly within the drought-affected subsample. Finally, refugee inflows influence migration dynamics by increasing demographic pressure in certain regions, suggesting that refugee presence may amplify internal migration through secondary displacement mechanisms.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 provides background information on Türkiye. Section 4 describes the econometric methodology. Section 5 presents the data and summary statistics. Section 6 outlines the identification strategy. Section 7 reports the empirical results. Section 8 discusses the findings, and Section 9 concludes.

[3]The analysis begins in 2008 due to the availability of consistent migration data from the Turkish Statistical Institute.

[4]The selection of these provinces is based on climatic exposure, geographic location in the southern part of Türkiye, and urban–rural characteristics.

[5]The Bagnouls–Gaussen index measures climatic dryness by comparing temperature and precipitation levels and is widely used to identify drought-prone regions.

2 Literature Review

Nordhaus (2019)[6] highlights that climate change has become particularly critical since the early 2000s. While natural climate variability has produced temperature anomalies over the last half-million years, recent decades have witnessed sustained warming of both land and oceans, primarily driven by human activity. This warming alters temperature extremes, precipitation patterns, storm activity, snowpacks, river runoff, water availability, and ice sheets. Climate models predict a significant rise in global temperatures by 2100, making the consequences of continued warming unavoidable without major mitigation efforts (Nordhaus and Boyer, 2003; Nordhaus, 2013). These long-run climatic trends provide the underlying context for a growing body of research examining how climate variability influences human mobility and migration decisions.

Figure 1 presents a reconstruction of global temperatures over the past half-million years based on Antarctic ice-core data. The series is normalized to the year 2000, set as the baseline of 0°C, and projections suggest that, without climate change mitigation policies, future temperatures will exceed the historical maximum of this period (Nordhaus, 2019).

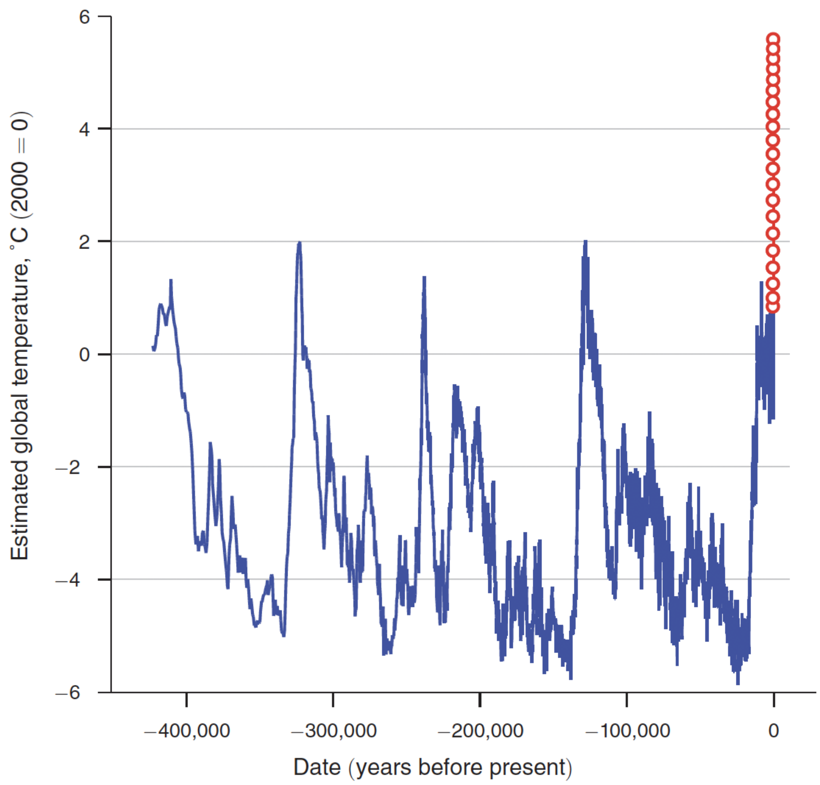

Figure 1: Estimated Global Temperature Changes Over the Past 400,000 Years

Note: Antarctic ice core data (solid line) and model projections for the next two centuries (circles). Data Source: Nordhaus (2013, 2019)

A growing body of empirical research examines the relationship between climate change and migration, documenting how climatic conditions influence human mobility across different contexts (Berlemann and Tran, 2020; Beine and Jeusette, 2021; Backhaus et al., 2015; Beine and Parsons, 2017; Berlemann and Steinhardt, 2017). A number of studies show that rising temperatures and changing precipitation patterns

[6] William Nordhaus was awarded the Nobel Prize in Economic Sciences in 2018 for integrating climate change into long-run macroeconomic analysis. His work has been pivotal in understanding the economic impacts of climate change and the importance of sustainable policies.

play a central role in shaping migration decisions (Backhaus et al., 2015; Cai et al., 2016; Cattaneo and Peri, 2016; Beine and Parsons, 2017; Berlemann and Steinhardt, 2017; Benveniste et al., 2022). These environmental changes increase the frequency and severity of natural hazards such as droughts, floods, sea-level rise, and wildfires, which in turn influence migration flows (Suhrke and Hazarika, 1993; Halliday, 2006; Millock, 2015; Report; IPCC, 2022).

At the same time, migration responses to climate variability are shaped by economic and social conditions. Classical and modern migration theories emphasize that mobility decisions depend on a combination of environmental pressures, economic opportunities, and institutional constraints (Ravenstein, 1885; Black, 2001; Cattaneo and Peri, 2016; Thorn et al., 2023). As Piguet et al. (2011) argue, climate change affects both the willingness and the ability to migrate, making it difficult to distinguish between direct and indirect effects. For instance, rising sea levels have rendered areas such as Tuvalu increasingly uninhabitable, forcing migration as a direct response to environmental change (Campbell and Warrick, 2014). In contrast, climate-induced declines in agricultural productivity and income represent indirect channels through which environmental change influences migration decisions (Coniglio and Pesce, 2015; Beine and Parsons, 2017).

Meta-analyses and comparative studies further highlight the heterogeneity of climate–migration relationships. Beine and Jeusette (2021) provide a comprehensive synthesis of empirical findings, emphasizing variation in both estimated effects and methodological approaches. A large empirical literature also documents how climatic shocks influence migration across different settings (Beine and Parsons, 2015; Millock, 2015; Ruiz et al., 2017; Berlemann and Steinhardt, 2017; Moore and Wesselbaum, 2023). In particular, studies such as Missirian and Schlenker (2017) and Carleton et al. (2016) show that temperature increases can significantly affect human behavior and welfare outcomes, including migration and mortality, with nonlinear responses to extreme heat.

Several studies identify systematic spatial patterns in climate-induced migration. Climate variability has been shown to drive migration from low- to high-altitude areas (Munshi, 2003; Moore and Wesselbaum, 2023). More broadly, the literature highlights a North–South divide: while countries in the Global North benefit from stronger institutions and greater adaptive capacity, regions in the Global South face higher exposure to climate risks, leading to migration toward wealthier and less climate-vulnerable destinations (Barrios et al., 2006; Kharin et al., 2007; Rappaport, 2009; Sasser, 2010; Hijzen and Wright, 2010; Cattaneo and Peri, 2016; Piguet et al., 2018). Projections underscore the magnitude of this phenomenon. Myers (2002) estimates that more than 200 million people could be displaced by climate change by 2050, while Rigaud et al. (2018) projects that over 143 million individuals may migrate internally within developing regions.

A substantial strand of the literature focuses on the role of specific climatic variables. Rising temperatures have been shown to significantly influence migration by directly affecting living conditions (Bohra-Mishra et al., 2014; Beine and Parsons, 2017). However, these effects are heterogeneous: in middle-income countries, higher temperatures tend to increase out-migration, whereas in poorer countries they may reduce migration due to liquidity constraints (Cattaneo and Peri, 2016). Precipitation also plays a complex role in shaping migration decisions. For example, typhoon-induced floods in Vietnam triggered substantial internal migration (Berlemann and Tran, 2020), while increased rainfall in Ecuador reduced both internal and international migration (Gray, 2009). Beyond these direct effects, climatic shocks that reduce agricultural productivity and household income influence migration indirectly by constraining or reshaping mobility decisions (Coniglio and Pesce, 2015; Cattaneo and Peri, 2016).

Climate change also affects migration through broader economic and social mechanisms. A large body of research shows that climatic shocks reduce agricultural production, increasing poverty and vulnerability (Cai et al., 2016; Cattaneo and Peri, 2016; Beine and Parsons, 2017). These pressures are intensified by extreme temperatures and recurrent droughts (Roberts et al., 2013), which can contribute to conflict and displacement (Koubi et al., 2016; Kelley et al., 2017; Selby et al., 2017; Chen and Mueller, 2019; Erdoğan and Cantürk, 2022). A prominent example is the Syrian drought, where Kelley et al. (2015) shows that climate conditions contributed to agricultural collapse and economic instability, ultimately intensifying conflict and generating large refugee flows. These dynamics highlight the secondary effects of climate change–induced migration, including increased pressure on host regions and heightened risks of social tension (Barrios et al., 2006; Joseph and Wodon, 2013; Strobl and Valfort, 2015; Thiede et al., 2016; Dallmann and Millock, 2017; Ruiz et al., 2017; Gray et al., 2020; Zouabi, 2021). Migration may also reshape cultural interactions between host and migrant populations through compositional changes and cultural diffusion (Rapoport et al., 2020).

Despite this extensive global literature, evidence for Türkiye remains limited. Existing studies have largely examined climate impacts and migration separately. On the climate side, Cebeci et al. (2019) analyzes drought vulnerability across Turkish cities, while government reports document the impacts of floods, wildfires, and droughts. Other studies examine environmental and urban dynamics in the context of climate change (Özceylan and Coşkun, 2012; Benli et al., 2018; Kiziroğlu, 2017). On the migration side, Erdoğan and Cantürk (2022) analyzes international migration flows to Türkiye. However, these studies do not systematically examine how climate variability shapes internal migration patterns. This gap motivates our analysis and highlights the contribution of this study.

Finally, this paper contributes to the literature in several important ways. First, it provides one of the first comprehensive analyses of internal migration in Türkiye that jointly considers climatic, economic, and demographic factors within a unified empirical framework. While existing studies primarily examine climate impacts or migration separately, this paper integrates both dimensions and explicitly accounts for conditions in both origin and destination locations.

Second, the paper contributes to the growing literature on climate-induced migration by documenting nonlinear temperature effects, showing that migration responses increase sharply beyond a critical threshold. This finding is consistent with recent evidence on the nonlinear impacts of extreme heat on human outcomes (Carleton et al., 2016; Missirian and Schlenker, 2017), and highlights the importance of accounting for temperature extremes rather than relying solely on average climate measures.

Third, by employing multiple estimation strategies, including a three-way fixed effects, PPML, and Hausman–Taylor, the analysis provides robust evidence across different econometric specifications, addressing key challenges such as zero migration flows, endogeneity, and unobserved heterogeneity.

Finally, the paper introduces refugee inflows as a secondary channel through which climate-related pressures influence internal migration. By showing that refugee presence can amplify migration through demographic and economic pressures, the analysis highlights an indirect mechanism linking climate change to population movements, which remains relatively underexplored in the literature.

3 Overview of Türkiye

Türkiye, situated between 36° and 42° north latitude and 26° and 45° east longitude, lies at the crossroads of Europe and Asia, exposing it to diverse climatic, geographic, and socio-economic conditions that make it particularly sensitive to the impacts of climate change. With a population of approximately 86 million and a relatively young median age of about 32 years, the country is highly urbanized, as nearly 75% of people reside in metropolitan areas such as Istanbul, Ankara, and Izmir, largely driven by rural to urban migration in search of employment, education, and improved living standards. Rural areas, however, remain dependent on agriculture and livestock, increasing their vulnerability to drought and extreme weather events. In addition, Türkiye hosts nearly 5 million refugees, of whom approximately 3.5 million are from Syria, reflecting its central role in regional migration dynamics.[7]

Türkiye's geography features sharp contrasts in topography and climate over relatively short distances, producing substantial regional disparities in socio-economic conditions. The country experiences three dominant climate types: Mediterranean in the southwest, Black Sea in the north, and Continental across the interior and eastern regions. Much of the Black Sea, Mediterranean, and Eastern Anatolia regions is mountainous, with east–west ranges reaching considerable elevations. This geographic configuration generates strong variations in climate, livelihoods, and lifestyles across the country (Tekin, 2023).

Türkiye is part of the Mediterranean Basin, one of the regions most vulnerable to climate change. Since the 1980s, the country has experienced notable shifts in temperature and precipitation, with a clear trend toward a warmer and drier climate (Türkeş and Tatlı, 2009; Türkes, 2012; Türkeş, 2017). Climate projections indicate that these patterns will intensify in the coming decades (Bayram and Öztürk, 2021; Bozkurt and Sen, 2013; Demircan et al., 2017). Türkiye's precipitation climatology is strongly shaped by the North Atlantic Oscillation (NAO): positive phases of the NAO are linked to severe droughts, while negative phases bring increased rainfall (Türkeş and Erlat, 2005). With climate change, the positive intensity of this oscillation—and thus the risk of drought in Türkiye—is expected to rise (Visbeck et al., 2001).

Across Türkiye, climate-related events have become increasingly evident in recent decades. Figure 2

[7] Official statistics available at: https://en.goc.gov.tr/

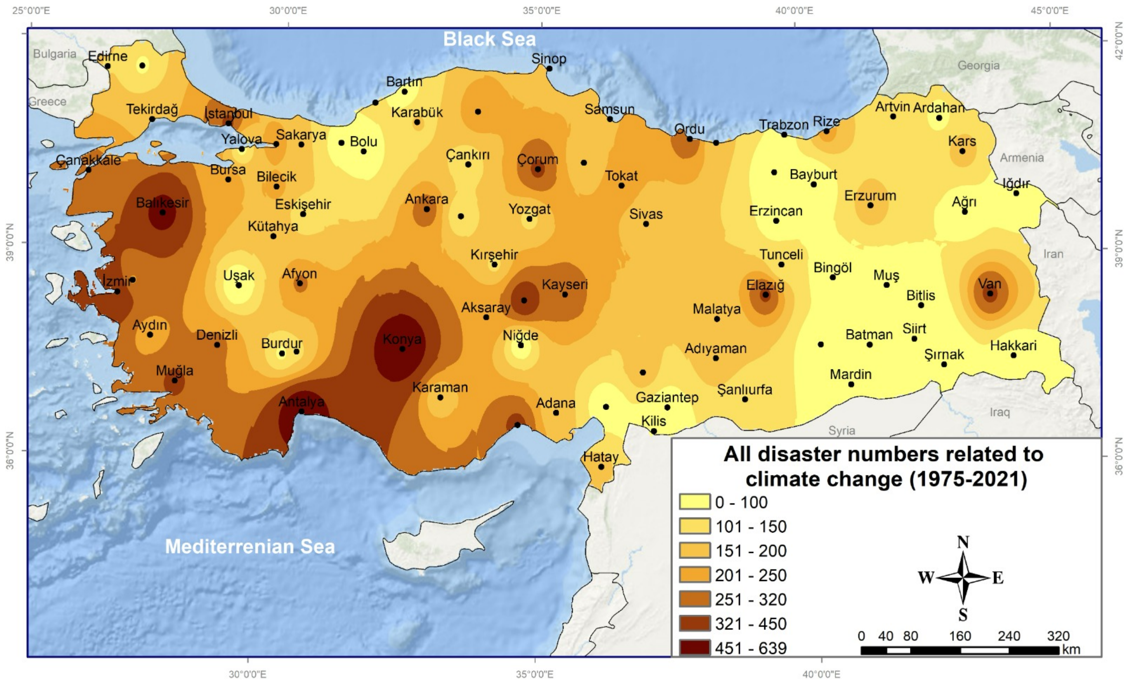

Figure 2: Total Climate Events in Türkiye from 1975 to 2021

Note: The figure shows the total number of climate-related disaster events, including droughts, extreme temperatures, heatwaves, floods, wildfires, and storms.
Data Source: Tekin (2023)

illustrates the frequency of such events between 1975 and 2021.[8] The southern regions appear particularly affected, reflecting both their geographic vulnerability to extreme weather and their concentration of major urban and economic centers. The visualization highlights the regional distribution of disasters, including droughts, extreme temperatures, heatwaves, floods, wildfires, and storms, with darker shades indicating higher frequencies of events.

Drought is poised to be one of the most significant climatic disasters shaping Türkiye's future (Tekin, 2023). Besides the Black Sea region, drought conditions are expected to intensify mainly in the Central Anatolia, Southeastern Anatolia, and Mediterranean regions. Current data indicate that a significant portion of Türkiye's land is at risk of drought and desertification (Türkeş, 2017; Bayram and Öztürk, 2021; Serkendiz et al., 2023). This situation threatens the primary livelihoods of rural populations, such as agriculture and livestock, potentially leading to increased migration.

Türkiye's position at the intersection of mid-latitude/polar and tropical-origin pressure systems leads to a variety of sub-climates. These physical and human factors contribute to high human mobility, making Türkiye one of the countries with the highest mobility rates globally. Historically, Türkiye has been a crucial transit route for migration. Many individuals from the Middle East and South Asia, such as Afghans, Iranians, Iraqis, Syrians, and Pakistanis, use Türkiye as a transit point to migrate to Europe. Recent years have seen an increase in natural disasters in these politically and economically turbulent regions. It is highly likely that Türkiye will continue to receive migrants from the Middle East and Western Asia, affected by drought and related issues, and from Southeast Asia, impacted by severe and irregular rainfall. Consequently, Türkiye is expected to face climate migrants from neighboring countries, as well as from Southeast Asia and Africa (Tekin, 2023). Studies indicate that many countries near Türkiye are at significant risk of food insecurity, heavily reliant on climate dependent livelihoods (Krishnamurthy et al., 2014). As climate change intensifies in these regions, it will likely trigger further migration.

[8]This figure reports only the number of recorded events and does not capture their intensity or consequences. Despite this limitation, it provides the most illustrative overview currently available.

4 Methodology

In this study, we use both the Gravity Model and the Random Utility Model to analyze the factors influencing migration. The Gravity Model in Economics was originally developed by (Tinbergen, 1962), then by (McCallum, 1995; Anderson and Van Wincoop, 2003) and then by (Rappaport, 2009) to explain international trade patterns and started to being used in labor and migration flows. Beine and Parsons (2015) adapted the Gravity Model from trade studies to migration. Afifi and Warner (2008); Bohra-Mishra et al. (2014); Backhaus et al. (2015); Beine and Parsons (2017) also used this model to analyze migration dynamics, taking into account geographic proximity, economic disparities, and the specific impact of climate variables like temperature and precipitation. The RUM establishes a mathematical framework to determine the factors influencing an individual's decision to migrate between cities within a country, assessing the utility of both the origin and destination cities based on current and expected future conditions (Beine and Parsons, 2015; Coniglio and Pesce, 2015; Beine and Parsons, 2017; Ruiz et al., 2017; Lewis and Swannell, 2018; Freeman and Lewis, 2021).

4.1 Random Utility Model

In this section, we establish a mathematical model using RUM (Beine et al., 2011; Beine and Parsons, 2017; Ruiz et al., 2017; Lewis and Swannell, 2018; Freeman and Lewis, 2021) to determine the conditions under which an individual decides to migrate from one city to another within a country. The RUM evaluates the utility derived from residing in the origin and destination cities, considering current utility and expected future conditions.

To define the utility function, we incorporate several factors that influence an individual's decision to migrate, broadly categorized as push and pull factors. As noted by Ravenstein (1889), people migrate to improve their circumstances, driven by social, political, and environmental conditions. Push factors are unfavorable conditions in the origin location, such as high living costs, adverse climate conditions, or limited economic opportunities, while pull factors refer to attractive features of the destination, such as better employment opportunities, favorable climate conditions, and lower living costs (Rummukainen, 2012).[9] Within this framework, Afifi and Warner (2008) emphasized that hunger, war, and violations of human rights function as extreme push factors leading to forced migration. Similarly, Burzyński et al. (2022) identified real wages, education levels, and migration costs as determinants that can either push individuals away from the origin or pull them toward a destination. Individual risk preferences also shape these dynamics: Becker (1962) argued that "risk-lovers" are more likely to migrate than "risk-averse" individuals, thereby amplifying the pull of potential opportunities. Finally, Rapoport et al. (2020) noted that migration entails not only economic push and pull forces but also cultural ones, as individuals weigh both the loss of familiar environments and the potential cultural opportunities offered by the destination.

An individual decides to migrate from origin i to destination j if the expected utility in the destination city exceeds the current utility in the origin city, as shown in equation (1). We prefer to use expectations for the next period while modeling our function, following the approach of (Reuveny and Moore, 2009; Coniglio and Pesce, 2015), who utilized expected values in their model. However, Cattaneo and Peri (2016); Beine and Parsons (2017) employed a two-period model with U_{t+1} for the next period, solving their models over two periods. Additionally, Rappaport (2009) modeled dynamic and steady-state levels over all periods. In addition, we do not consider interest rates and discount rates as Reuveny and Moore (2009) used in their model.

$$U_{it} < U^e_{jt} \tag{1}$$

The utility in origin city i at time t, U_{it}, is influenced by several factors, including income, private benefits such as family ties or attachment to the origin city, daily living costs, and climate change-related costs such as extreme temperatures and droughts.

First, let us express the utility function in general terms for the origin city in equation (2):

[9] Migration decisions are influenced by a wide range of factors, including demographic trends, policy interventions, and personal circumstances. For tractability, however, we restrict our model to the most widely studied and quantifiable determinants: economic, climatic, and social push and pull factors.

$$U_{it}(I_{it}, B_{it}, C_{it}) = I_{it} + B_{it} - C_{it} + \epsilon_i t \tag{2}$$

where I_{it} represents the income, B_{it} denotes the non-income (non-pecuniary) benefits of residing in the origin city, such as family ties, social networks, or attachment to place, and C_{it} indicates the costs associated with residing in the origin city. Finally, ϵ_{it} represents the unobserved (idiosyncratic) component of utility, capturing individual-specific preferences and factors not directly measurable within the model.

Next, let us define the components I_{it}, B_{it}, and C_{it} of U_{it}:

Income in the origin city is given by:

$$I_{it} = W_{it} + PI_{it} \tag{3}$$

where W_{it} denotes wealth, including assets such as houses, and PI_{it} represents permanent income, such as salary or wages.[10] Beine et al. (2011) mentioned wages as income, but in our model, we prefer to use personal income.

Benefit in the origin city is given by:

$$B_{it} = PB_{it} \tag{4}$$

where PB_{it} denotes personal benefits such as family ties, friendships, habits, and attachments to the local area. We assume that the individuals in the origin city are risk-averse and homophilic (Rapoport et al., 2020), preferring not to change their location or social circle; thus, B_it is an increasing function. As observed in Turkish culture, people often have a strong attachment to their place of residence and lifestyle.

Costs in the origin city is given by:

$$C_{it} = DC_{it} + RB_{it} + CCC_{it} \tag{5}$$

where DC_{it} denotes daily costs which is the costs for living, RB represents refugee burden, and CCC_{it} denotes climate change costs. Beine and Parsons (2017) put their model climate change as a non-pecuniary cost. Given the focus of this paper on the impact of climate change, CCC_{it} can be further defined as:

$$CCC_{it} = T_{it} - P_{it} \tag{6}$$

where T_{it} represents the temperature and P_{it} denotes precipitation in the origin city i at time t. Temperature is an increasing function beyond a certain threshold, and we assume that an increase in temperature incurs additional costs with extreme heat being detrimental. Conversely, precipitation has a cost-saving effect, which can be beneficial against droughts but may also lead to negative utility in certain regions of Türkiye since too much rain is not desirable.

Thus, we can rewrite the utility function for the origin as in equation (7):

$$U_{it} = W_{it} + PI_{it} + PB_{it} - DC_{it} - RB_{it} - T_{it} + P_{it} \tag{7}$$

According to (7), the utility of a person in the origin city generally depends on wealth, personal income, personal benefits from living in the origin city, the presence of refugees, and the impact of temperature and precipitation.

In the case of Türkiye, an increase in temperature, especially during the summer, will lead people to migrate. Bohra-Mishra et al. (2014) stated that temperature has a nonlinear effect with a threshold where the average monthly temperature above 25°C significantly impacts migration decisions. Burzyński et al. (2022) noted that 25°C leads to lower productivity. Burke et al. (2015) suggested a threshold between 20°C and 30°C for migration. Given the occurrence of extreme temperatures in Türkiye especially in summer, living in those areas becomes costly, prompting people to relocate. An increase in precipitation is beneficial for individuals, especially those in the origin city engaged in agriculture. Barrios et al. (2006) modeled the relationship between agricultural production and precipitation. In addition, there are more than 5 million refugees living in Türkiye, a situation that has persisted for a long time. In some cities, the local population

[10] This concept is derived from John Maynard Keynes' hypothesis on the Permanent Income Theory, which classifies income into wealth and permanent income. We assume people who decide to move have only one house as wealth and cannot sell or rent this house because of family heritage.

is very unsatisfied with this situation. The presence of these refugees creates negative externalities in terms of education and health services. Therefore, we include refugee burden in our model as a parameter in the cost function.

Similarly, expected the utility in the destination j at time t, U^e_{jt}, is influenced by many factors such as income, better climate conditions, migration costs from the origin to the destination, daily costs of living in the destination, and positive factors such as new opportunities and better expectations or hopes.

First, let us express the utility function for the destination city as expected values:

$$U^e_{jt}(I_{jt}, B_{jt}, C_{jt}) = I^e_{jt} + B^e_{jt} - C^e_{jt} + \epsilon_{jt} \tag{8}$$

where $\epsilon_j t$ represents the unobserved (idiosyncratic) component of utility, capturing individual-specific preferences and factors not directly measurable within the model.

Next, let us define the components of U^e_{jt}:

Income in the destination city is given by:

$$I^e jt = PI^e_{jt} \tag{9}$$

where PI^e_{jt} is the expected personal income in destination city j at time t. We did not put wealth in equation (9) because we assumed that a person who lives in the origin city can not sell or rent his house because of heritage issues.

Benefits in the destination city are defined as:

$$B^e_{jt} = BCC^e_{jt} + BH^e_{jt} \tag{10}$$

where BCC^e_{jt} represents better climate conditions in destination city j at time t, BH^e_{jt} represents expected benefits of quality of life as (Rappaport, 2009) discussed. Also, if a person is a risk-lover, he will expect more benefits from migration (Becker, 1962).

Costs in the destination city are expressed as:

$$C^e_{jt} = MC^e_{ij} + D_{ij} + DC^e_{jt} \tag{11}$$

where MC^e_{ij} is the migration cost from origin city i to destination city j, D_{ij} is the distance between origin city i and destination city j, and DC^e_{jt} is the daily cost of living in destination city j at time t. We prefer to use travel costs as distance and think the cost is 1 unit per distance. Anderson (2013); Santana-Gallego and Paniagua (2022) mentioned the iceberg cost, but we prefer to use migration cost.

By combining the expected values of the functions, the destination's utility function is written as follows:

$$U^e_{jt} = PI^e_{jt} + BCC^e_{jt} + BH^e_{jt} - MC^e_{ij} - D_{ij} - DC^e_{jt} \tag{12}$$

Equation (12) indicates that the utility of a person considering migration depends on the expected personal income in the destination city, better climate conditions, benefits of hope possibly as being risk-lover, migration cost, distance (which is directly considered as a cost), and daily cost.

So, the decision to migrate can be expressed as:

$$W_{it} + PI_{it} + PB_{it} - DC_{it} - RB_{it} - T_{it} + P_{it} < PI^e_{jt} + BCC^e_{jt} + BH^e_{jt} - MC^e_{ij} - D_{ij} - DC^e_{jt} \tag{13}$$

For simplicity , we assume PI_{it} and PI^e_{jt} are equal; DC_{it} and DC^e_{jt} are equal; PB_{it} and BH^e_{jt} are equal. The new equation will be:

$$W_{it} + PB_{it} - RB_{it} - T_{it} + P_{it} < BCC^e_{jt} - MC^e_{ij} - D_{ij} \tag{14}$$

Equation (14) captures the essential elements influencing an individual's migration decision by comparing the utility derived from the factors in both the origin and destination cities. Specifically, it shows that if the total utility of a person in the origin city exceeds that in the destination city, the person will be inclined to migrate.

According to equation (14), we can conclude that climate change will affect personal preferences. If the temperature increases, people seek new places in response to the hot temperatures. We did not include temperature as a cost in the destination city since individuals migrate with the expectation of better climate conditions and favorable weather, as suggested by Rappaport (2009). Additionally, we considered the refugee burden as a cost since the capacity of a place depends on the number of people. Refugee burden also causes conflicts in cultural aspects, resulting in negative externalities and increased costs for the local population.

Let's proceed to calculate the migration probability function. The probability that $U_{jt}^e > U_{it}$ can be represented using an exponential function.

$$P(U_{jt}^e > U_{it}) = \min\left(1, \frac{e^{U_{jt}^e}}{e^{U_{it}}}\right) \tag{15}$$

Substituting the utility functions, we get:

$$P(U_{jt}^e > U_{it}) = \min\left(1, \frac{e^{BCC_{jt}^e - TMC_{ij}^e - D_{ij}}}{e^{W_{it} + PB_{it} - RB_{it} - T_{it} + P_{it}}}\right) \tag{16}$$

Now, let's consider the probability of choosing a particular destination j among all possible destinations j. Assuming there are multiple possible destinations, the individual will choose the destination that maximizes his utility. The probability of choosing destination j out of all possible destinations j is given by:

$$P(\text{Choose } j \mid \text{Migration}) = \frac{e^{U_{jt}^e}}{\sum_{k \in J} e^{U_{kt}^e}} \tag{17}$$

We can combine the the equation (16) and (17) and get equation (18) :

Combining these two steps, the overall probability of migrating from origin i to destination j is:

$$P(\text{Migration to } j) = \min\left(1, \frac{e^{BCC_{jt}^e - MC_{ij}^e - D_{ij}}}{e^{W_{it} + PB_{it} - RB_{it} - T_{it} + P_{it}}}\right) \cdot \left(\frac{e^{BCC_{jt}^e - MC_{ij}^e - D_{ij}}}{\sum_{k \in J} e^{BCC_{kt}^e - MC_{ik}^e - D_{ik}^e}}\right) \tag{18}$$

The equation (18) tells us that the decision of a person for migration depends on two main components. First, the expected utility in the destination city should be higher than the current utility in the origin city. Second, among all possible destination options, the individual will choose the one that offers the highest probability of better utility. Thus, the highest probability destination will be selected for migration.

According to Equation (18), migration from an origin to a destination city is primarily influenced by factors such as temperature, precipitation, distance, income, and migration cost. If the temperature increases, precipitation decreases, distance increases, migration cost increases, and income decreases, the probability of migration will decrease. Conversely, if the temperature decreases, precipitation increases, distance decreases, migration cost decreases, and income increases, the probability of migration will increase.

we can rewrite the equation as:

$$M_{ijt} = \frac{e^{BCC_{jt}^e - MC_{ij}^e - D_{ij}}}{e^{W_{it} + PB_{it} - RB_{it} - T_{it} + P_{it}}} \cdot \eta_{ijt} \tag{19}$$

where η_{ijt} represents the unobserved (idiosyncratic) component of utility, capturing individual-specific preferences and factors not directly measurable within the model.

4.2 Gravity Model

The gravity model aims to formalize and predict the geography of flows or interactions.[11] It is commonly employed to predict trade flows, but recently it has also been applied to migration trends.(Anderson and Van Wincoop, 2003; Afifi and Warner, 2008; Beine et al., 2011; Beine and Parsons, 2017; Grohmann, 2023).

The gravity model of international trade is a foundational tool in economics for analyzing trade flows between countries. The basic form of the gravity model posits that trade between two countries is proportional

[11]The law posits that the attractive force between two objects i and j is given by: $F = G\frac{M_i \times M_j}{D_{ij}^2}$, where F represents the attraction force, M_i and M_j are the respective masses of the objects, D_{ij} is the distance between them, and G is the gravitational constant, which depends on the units of measurement for mass and force.

to the product of their economic sizes (typically GDP) and inversely proportional to the distance between them.

The simplest form of the gravity equation, as introduced by Tinbergen (1962) and widely used by (McCallum, 1995; Anderson and Van Wincoop, 2003; Silva and Tenreyro, 2006a; Anderson, 2013; Rapoport et al., 2020), can be expressed as:

$$T_{ij} = \alpha \frac{Y_i Y_j}{D_{ij}} \tag{20}$$

where T_{ij} is the trade flow from country i to country j, Y_i and Y_j are the GDPs of countries i and j, respectively, D_{ij} is the distance between the two countries, and α is a constant.

McCallum (1995) used this model to illustrate the significant impact of national borders on trade, famously showing that Canadian provinces trade more with each other than with U.S. states, even when controlling for distance and economic size.

Anderson and Van Wincoop (2003); Yotov (2024); Anderson and Yotov (2025) refined the model by introducing multilateral resistance terms to account for the fact that trade between two countries depends not only on their bilateral barriers but also on their trade barriers with other countries. The modified equation is:

$$T_{ij} = \frac{Y_i Y_j}{Y_w} \left(\frac{t_{ij}}{P_i P_j} \right)^{1-\sigma} \tag{21}$$

where Y_w is the world GDP, t_{ij} is the bilateral trade cost between countries i and j, P_i and P_j are the multilateral resistance terms, and σ is the elasticity of substitution. This framework highlights the importance of considering the relative resistance to trade, not just the bilateral distance.

Silva and Tenreyro (2006a) addressed the issue of zero trade flows, which pose a problem for log-linearized models. They proposed using PPML estimator to handle zero trade values and heteroskedasticity. Then the model becomes:

$$E[T_{ij}] = \exp(\beta_0 + \beta_1 \ln Y_i + \beta_2 \ln Y_j + \beta_3 \ln D_{ij} + \mathbf{X}_{ij}'\beta) \tag{22}$$

where $\mathbf{X}_{ij}$ represents additional independent control variables.

Rappaport (2009); Santana-Gallego and Paniagua (2022) extended the gravity model by incorporating additional variables that influence trade, such as common language, colonial history, and regional trade agreements. The extended model is:

$$T_{ij} = \alpha \left(\frac{Y_i^{\beta_1} Y_j^{\beta_2}}{D_{ij}^{\beta_3}} \right) \exp(\gamma_1 \text{Lang}_{ij} + \gamma_2 \text{Col}_{ij} + \gamma_3 \text{RTA}_{ij} + u_{ij}) \tag{23}$$

where Lang_{ij} is a dummy variable for common language, Col_{ij} is a dummy variable for colonial ties, RTA_{ij} is a dummy variable for regional trade agreements, and u_{ij} is an error term (Orefice, 2015).

Combining insights from Anderson (2013) and Silva and Tenreyro (2006a), we can present the gravity model in a functional form that addresses heteroskedasticity and non-linearity:

$$T_{ij} = \alpha Y_i^{\beta_1} Y_j^{\beta_2} D_{ij}^{\beta_3} \exp(\gamma \mathbf{Z}_{ij}) \epsilon_{ij} \tag{24}$$

where $\mathbf{Z}_{ij}$ includes all additional explanatory variables, and ϵ_{ij} represents the unobserved (idiosyncratic) component of trade flows, capturing individual-specific preferences and factors not directly measurable within the model.

In the context of migration, the same functional form can be adapted by replacing trade flows with migration flows and explicitly incorporating economic and climatic variables as the main determinants. Thus, the gravity equation becomes:

$$M_{ij} = \alpha \, GDP_i^{\beta_1} \, GDP_j^{\beta_2} \, D_{ij}^{\beta_3} \, \exp(\gamma_1 Temp_i + \gamma_2 Temp_j + \gamma_3 Prec_i + \gamma_4 Prec_j + \gamma_5 RB_i) \, \epsilon_{ij} \tag{25}$$

where M_{ij} is the migration flow from origin i to destination j, GDP_i and GDP_j represent the economic size of the origin and destination, D_{ij} is the distance between them, $Temp_i$ and $Temp_j$ are the average temperatures, $Prec_i$ and $Prec_j$ are the levels of precipitation, and RB_i denotes the refugee burden at the origin. ϵ_{ij} represents the unobserved component of migration, capturing individual-specific preferences and factors not directly measurable within the model.

4.3 Econometric Specification

To empirically examine the determinants of internal migration in Türkiye, we translate the theoretical framework developed in the previous sections into an econometric specification. The empirical strategy follows a gravity-type modeling approach that is grounded in the Random Utility Model (RUM) and widely used in migration studies. This framework explains migration flows between origin and destination provinces as a function of climatic conditions, economic characteristics, and spatial frictions.

In the baseline specification, migration flows are modeled using a gravity-based migration equation that incorporates climatic, economic, and geographic determinants. We then extend this specification to allow for nonlinear responses of migration to environmental conditions, particularly temperature and precipitation, which may generate stronger migration responses under extreme climatic stress.

4.3.1 Gravity-Based Migration Model

The empirical specification combines insights from the Random Utility Model (RUM) and the gravity framework to analyze the determinants of internal migration flows. This integrated approach allows us to link individual migration decisions, which are based on utility maximization, with the broader structural factors that shape spatial interactions between regions.

RUM provides the microeconomic foundation for migration behavior, suggesting that individuals choose to migrate when the expected utility in a destination location exceeds the utility of remaining in the origin location. These decisions are influenced by economic opportunities, environmental conditions, and migration costs (Beine and Parsons, 2015, 2017; Ruiz et al., 2017).

The gravity model complements this framework by capturing the spatial structure of migration flows. Originally developed in the context of international trade by Tinbergen (1962) and later refined by Anderson and Van Wincoop (2003), the gravity model predicts that interactions between locations increase with economic size and decrease with distance. This framework has been widely applied to migration studies (Beine and Parsons, 2015; Backhaus et al., 2015; Beine and Parsons, 2017).

Combining these two frameworks provides a natural way to model migration flows influenced by climatic conditions, economic characteristics, and spatial frictions. Following the empirical approaches used in (Backhaus et al., 2015; Beine and Parsons, 2015, 2017; Ruiz et al., 2017), the resulting empirical specification can be expressed as:

$$\begin{aligned} \ln(M_{ijt}) = \alpha &+ \beta_1 \ln(T_{it}) + \beta_2 \ln(T_{jt}) + \beta_3 \ln(P_{it}) + \beta_4 \ln(P_{jt}) \\ &+ \beta_5 \ln(GDP_{it}) + \beta_6 \ln(GDP_{jt}) + \beta_7 \ln(D_{ij}) + \beta_8 \ln(RB_{it}) \\ &+ \lambda_i + \theta_j + \delta_t + \phi_{ij} + \epsilon_{ijt} \end{aligned} \tag{26}$$

where M_{ijt} denotes the migration flow from origin i to destination j in year t. The variables T_{it} and T_{jt} represent temperatures in the origin and destination locations, respectively, while P_{it} and P_{jt} denote precipitation levels. GDP_{it} and GDP_{jt} capture economic conditions in the origin and destination provinces. D_{ij} represents the geographic distance between the origin and destination provinces and proxies migration costs. RB_{it} denotes the refugee burden in the origin province.[12]

The model includes several layers of fixed effects to control for unobserved heterogeneity. The term λ_i captures time-invariant characteristics of origin provinces, θ_j captures destination-specific characteristics, and δ_t denotes year fixed effects controlling for national shocks that affect migration patterns over time. In addition, ϕ_{ij} represents origin–destination pair fixed effects that account for persistent bilateral migration patterns between provinces. The error term ϵ_{ijt} captures unobserved factors affecting migration flows.

For parsimony, the specification can be written in a more compact form as:

$$\ln(M_{ijt}) = \beta_0 + \beta_i \ln(X_{it}) + \beta_j \ln(X_{jt}) + \beta_{ij} \ln(D_{ij}) + \lambda_i + \theta_j + \delta_t + \phi_{ij} + \epsilon_{ijt} \tag{27}$$

[12] Equation (26) is presented in log-linear form for expositional clarity. In the empirical analysis, we estimate the fixed effects and Hausman-Taylor models using the logarithm of migration flows as the dependent variable. However, when applying the Poisson Pseudo-Maximum Likelihood (PPML) estimator, we use migration flows in levels as the dependent variable. This approach follows Silva and Tenreyro (2006b), who show that PPML provides consistent estimates for gravity-type models in the presence of heteroskedasticity and zero flows. Independent variables may still enter the PPML specification in logarithmic form.

where X_{it} and X_{jt} denote vectors of explanatory variables associated with the origin and destination provinces, respectively.

4.3.2 Nonlinear Climate Effects

While the baseline specification assumes a linear relationship between climatic variables and migration flows, a growing body of literature suggests that the effects of environmental conditions on migration are often nonlinear. In particular, moderate climatic variations may have limited effects on migration, whereas extreme temperatures or severe drought conditions can generate significantly stronger migration responses (Bohra-Mishra et al., 2014; Burke et al., 2015; Hsiao, 2021).

To capture these potential threshold effects, we extend the baseline model by including quadratic terms for temperature and precipitation. The nonlinear specification is given by:

$$\ln(M_{ijt}) = \beta_1 \ln(T_{it}) + \beta_2(\ln T_{it})^2 + \beta_3 \ln(P_{it}) + \beta_4(\ln P_{it})^2 + \sum \gamma Z_{it} + \theta \ln(D_{ij}) + \lambda_i + \theta_j + \delta_t + \epsilon_{ijt} \quad (28)$$

In this specification, the quadratic terms $(\ln T_{it})^2$ and $(\ln P_{it})^2$ allow the model to capture nonlinear responses of migration to climatic conditions. These terms enable the identification of threshold effects, in which extreme environmental conditions elicit stronger migration responses than moderate climatic changes.

The vector Z_{it} includes additional time-varying control variables measured at the origin province level, such as wildfire activity, electricity consumption as a proxy for adaptation infrastructure, and the share of Syrian refugees. The specification maintains the inclusion of origin, destination, and year fixed effects to control for unobserved spatial heterogeneity and common macroeconomic shocks affecting migration patterns over time.

4.4 Estimation Methods

To estimate the gravity-based migration model, we employ three complementary estimation approaches. First, we estimate the model using a three-way fixed effects specification to control for unobserved heterogeneity across origin provinces, destination provinces, and time periods. Second, we apply the Poisson Pseudo-Maximum Likelihood (PPML) estimator, which has become the standard method for estimating gravity-type models because it accounts for zero flows and heteroskedasticity. Finally, we employ the Hausman–Taylor estimator to address potential endogeneity concerns while allowing the estimation of time-invariant variables such as geographic distance.

Together, these estimation strategies allow us to assess the robustness of our results and ensure that the estimated relationships between climatic variables and migration flows are not driven by model-specific assumptions.

4.4.1 Three-Way Fixed Effects

We begin by estimating the model using a three-way fixed effects specification. This approach controls for unobserved heterogeneity across origin provinces, destination provinces, and time periods by including origin fixed effects, destination fixed effects, and year fixed effects in the regression.

Including these fixed effects allows us to control for time-invariant characteristics of provinces that may influence migration decisions, such as geographic conditions, long-term economic structure, or cultural factors. Year fixed effects capture common macroeconomic or national shocks affecting migration patterns over time. This specification is widely used in migration and gravity models to account for unobserved spatial heterogeneity and improve the consistency of estimated coefficients.

4.4.2 Poisson Pseudo-Maximum Likelihood Method

We also estimate the migration model using the Poisson Pseudo-Maximum Likelihood (PPML) estimator. Silva and Tenreyro (2006b) show that PPML provides consistent estimates for gravity-type models even when the dependent variable contains zeros, a common feature of migration and trade data.

Unlike log-linearized ordinary least squares estimations, PPML estimates the model using the dependent variable in levels. This approach avoids biases arising from heteroskedasticity and allows zero migration flows to be included in the estimation (Silva and Tenreyro, 2011; Linders and De Groot, 2006).

As a result, PPML has become the preferred estimator for gravity models in both trade and migration studies (Silva and Tenreyro, 2006b, 2011). The method has been widely applied in migration research (Beine et al., 2016; Beine and Parsons, 2017). For example, Ruiz et al. (2017) applied PPML to study the effect of climatic shocks on internal migration in Mexico, demonstrating its suitability for analyzing migration flows.

4.4.3 Hausman-Taylor Method

We also apply HT estimator to address endogeneity and unobserved heterogeneity in the migration model. Traditional fixed effects models control for unobserved heterogeneity but cannot estimate coefficients for time-invariant variables such as distance. Random effects models allow for time-invariant regressors but yield biased estimates when regressors are correlated with individual-specific effects. The HT approach overcomes these issues by decomposing the error term into individual-specific and idiosyncratic components, using a mixed-effects framework to address endogeneity (Hausman and Taylor, 1981; Wooldridge, 2010).

The HT estimator provides consistent estimates for both time-varying and time-invariant variables by exploiting variation within and between entities (Wooldridge, 2010). This makes it suitable for migration studies, where variables such as distance remain constant while others vary over time.

Empirical applications confirm its effectiveness. Egger and Pfaffermayr (2004) showed that the HT model addresses endogeneity in trade and FDI analysis, while Serlenga and Shin (2004) extended it to account for unobserved time-specific factors. Baltagi et al. (2003) demonstrated that the HT estimator exploits within and between variations of strictly exogenous variables as instruments, allowing consistent estimation when regressors correlate with individual effects. In this study, we apply the HT model to migration flows, providing one of the first applications of this method in this context.

5 Data

This section describes the data used to analyze internal migration flows in Türkiye. We outline the construction of the origin–destination panel and summarize the key climatic, economic, and demographic variables employed in the empirical analysis.

Administratively, Türkiye is divided into 81 provinces (*iller*), which are commonly referred to as cities in official statistics and form the basis of our dataset.[13] Using these administrative units, we construct an origin–destination panel that includes migration flows between all province pairs over the sample period. In addition to the full sample covering all 81 provinces, we also consider subsamples consisting of 30 provinces that are most exposed to climatic risks and a subset of 17 provinces identified as drought-affected, which are used in complementary analyses.

Climate-related factors play a central role in the construction of subsamples used in the analysis. In particular, we identify groups of provinces based on exposure to climatic stressors such as droughts, extreme heat, wildfires, and floods. Provinces that exceed predefined threshold levels in these indicators, including measures derived from drought indices, are classified as climate-exposed. Using this approach, we define a subsample of 30 provinces that are most affected by climatic conditions, as well as a narrower subsample of 17 provinces identified as drought-affected. These subsamples are used to complement the analysis based on the full sample of provinces.

The dataset covers the period from 2008 to 2024. Using all 81 provinces, we construct a balanced origin–destination–year panel of internal migration flows. Excluding self-migration, each year includes $81 \times 80 =$ 6,480 ordered province pairs, yielding a total of $6{,}480 \times 17 = 110{,}160$ observations. Subsample analyses based on climate-exposed provinces are constructed from this full panel and therefore contain fewer observations.

To illustrate the structure of the migration framework, consider a simplified example with two origin cities, labeled A and B, and five destination cities, labeled A, B, C, D, and E, as shown in Figure 3. In

[13]In Türkiye, the administrative unit officially called a "province" (*il*) is often translated into English as a "city." Each province contains multiple districts, towns, and villages; however, in statistical and economic analyses, the province is treated as the unit of observation. Accordingly, when we refer to "cities" in this study, we mean provinces at the Nomenclature of Territorial Units for Statistics-3 (NUTS-3) level.

this setting, each origin city can send migrants to all destination cities except itself, reflecting the restriction against self-migration. For example, city A can send migrants to cities B, C, D, and E, but not to itself, while city B can send migrants to cities A, C, D, and E. This simplified illustration generalizes to the full dataset, where all 81 provinces are treated symmetrically. In each year, every province can send migrants to the remaining 80 provinces, resulting in a complete set of ordered origin–destination pairs, excluding self-migration.

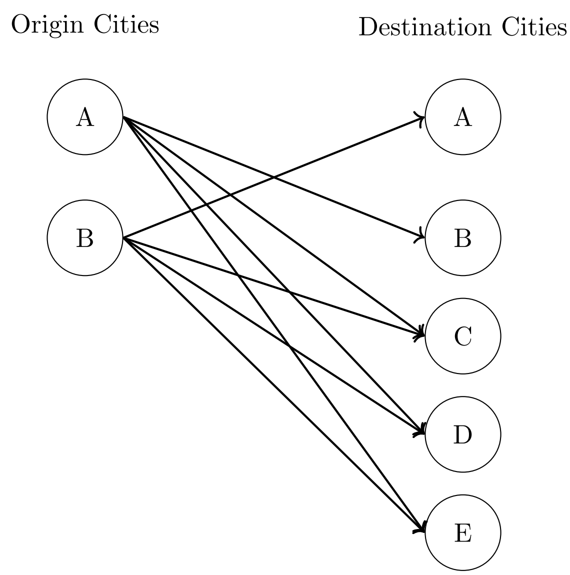

Figure 3: Illustration of Origin–Destination Migration Links

Note: The figure presents a simplified example with two origin cities (A, B) and five potential destination cities (A, B, C, D, E). Arrows indicate allowed origin–destination flows under the restriction against self-migration. The same structure applies in the empirical analysis, where each province can send migrants to all other provinces in a given year.

In the full sample, migration flows are modeled as unidirectional movements from origin to destination, with no restrictions imposed on migration between specific provinces beyond the exclusion of self-migration. This general structure is maintained when focusing on climate-related subsamples, including the 30 most climate-exposed provinces and the 17 drought-affected provinces. While migration patterns in these subsamples more closely reflect south-to-north and less-developed-to-more-developed movements commonly discussed in the climate migration literature, migration flows are still allowed between all province pairs within the relevant sample, ensuring consistency between the full-sample and subsample analyses.[14]

Figure 4 illustrates the spatial distribution of major climatic stressors across provinces in Türkiye and provides the empirical basis for the construction of climate-related subsamples used in the analysis. The map highlights substantial regional heterogeneity in exposure to droughts, floods, and wildfires. Drought exposure is widespread across much of the country, flood risk is more pronounced in northern provinces, and wildfire activity is concentrated mainly in southern regions. These spatial patterns reflect differences in geographic location and climatic conditions documented in the literature (Burzyński et al., 2022). While the main empirical analysis includes all 81 provinces, the figure is used to identify provinces that are more severely affected by adverse climatic conditions. In particular, provinces with relatively high exposure to these stressors form a subsample of 30 climate-exposed provinces, and a narrower subsample of 17 drought-affected provinces. These subsamples are employed in complementary analyses to assess whether migration responses differ in areas facing more intense climatic pressures, without restricting the full-sample migration framework.

[14]Beine and Parsons (2015) document differences in migration determinants between movements toward the global North and South, highlighting the role of economic incentives, skill composition, and distance.

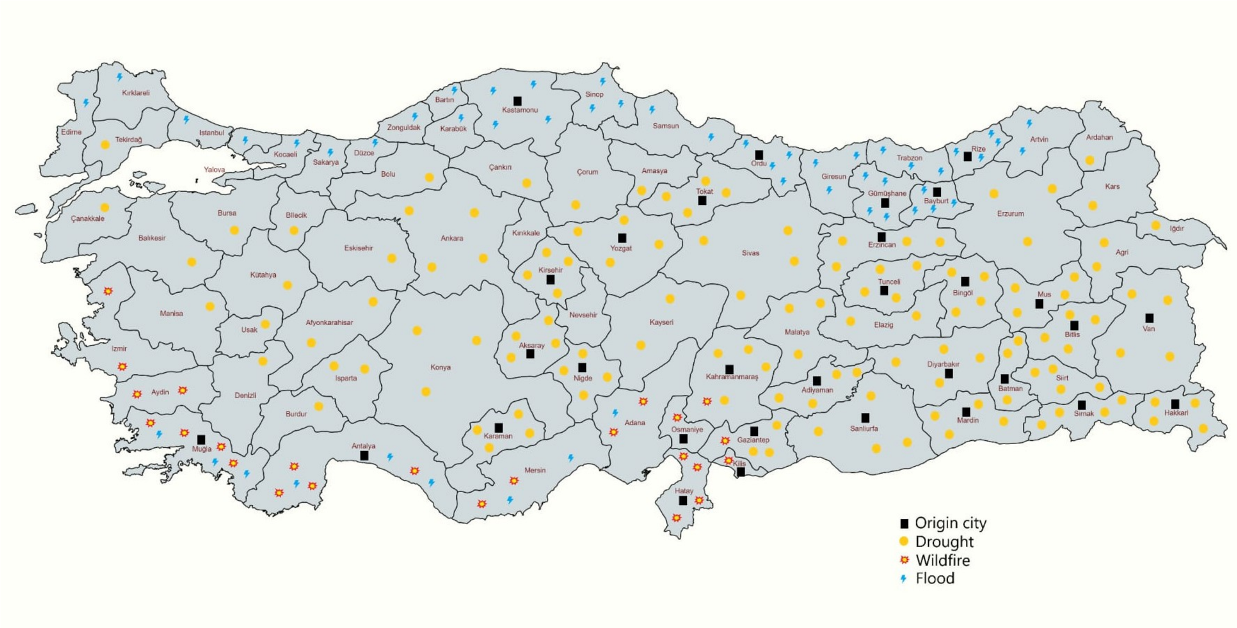

Figure 4: Spatial Distribution of Climatic Exposure across Provinces in Türkiye

Note: The figure illustrates the spatial distribution of major climatic stressors across provinces in Türkiye. Provinces affected by drought conditions are identified using the Bagnouls–Gaussen aridity index and are marked in yellow. Provinces primarily exposed to floods are indicated in blue, while provinces experiencing significant wildfire activity are shown in red with yellow interiors. Black squares denote provinces included in the climate-exposed subsamples used in the analysis. Drought exposure is widespread across the country, flood exposure is more prevalent in northern regions, and wildfire activity is concentrated mainly in southern provinces. *Data sources:* Turkish State Meteorological Service; https://www.mapchart.net/turkiye.html/.

Table 1 reports summary statistics for all variables used in the analysis, including the number of observations, means, standard deviations, and ranges. Most variables are observed for the full sample period, yielding 110,160 observations. However, several origin-level socioeconomic variables have fewer observations due to differences in data availability over time. In particular, information on Syrian refugees and crop production values is available only for a shorter subperiod, while data on fertility rates, hospitals, students, and students per classroom are reported only for selected years. These differences in coverage explain the variation in the number of observations across variables.

Migration data were obtained from the Turkish Statistical Institute (TSI) Population Statistics Portal.[15] Migration flows are reported annually at the province level and record movements from origin to destination provinces. During data preparation, province names were carefully standardized to avoid inconsistencies in sorting caused by Turkish characters. Zero migration flows between some origin–destination pairs are observed in the data and are retained in the analysis. Migration flows range from zero to approximately 30,000 individuals.

Monthly temperature and precipitation data were obtained from the Turkish State Meteorological Service (TSMS).[16] Using these monthly records, annual climate variables were constructed by calculating yearly averages for temperature and annual totals for precipitation. In addition to these aggregate measures, seasonal indicators were constructed to capture temperature extremes, including summer maximum temperatures and winter minimum temperatures. All climate variables were constructed separately for origin and destination provinces, allowing for distinct push and pull effects of climatic conditions in migration decisions. Across provinces and years, average annual temperatures range from approximately 4°C in colder regions to about 22°C in warmer areas, while annual precipitation ranges from roughly 120 mm to 3,100 mm. Seasonal

[15]Data source: https://nip.tuik.gov.tr

[16]Data source: https://www.mgm.gov.tr/eng

measures indicate substantial variation in both summer heat and winter cold.

Table 1: Summary Statistics

	Obs.	Mean	Std. Dev.	Min	Max
Migration flow (origin to destination)	110,160	406	1,208	0	30,036
Distance (km)	110,160	762	395	30	2,016
Origin characteristics					
Average temperature (°C)	110,160	14.6	3.3	4.1	22.1
Mean maximum temperature (°C)	110,160	20.2	3.0	10.5	28.1
Mean minimum temperature (°C)	110,160	9.1	3.9	−2.8	18.6
Winter minimum temperature (°C)	110,160	0.6	5.0	−20.5	10.9
Summer maximum temperature (°C)	110,160	31.1	3.4	21.1	40.1
Total precipitation (mm)	110,160	643	322	121	3,097
Winter precipitation (mm)	110,160	229	158	6	1,083
Summer precipitation (mm)	110,160	87	91	0	868
GDP (thousand TL)	110,160	91,518,579	486,800,000	568,507	13,000,000,000
GDP per capita (USD)	110,160	8,195	3,498	2,713	24,452
GDP per capita (TL)	110,160	65,239	104,615	3,888	802,669
Electricity consumption per capita (kWh)	110,160	2,573	1,670	443	11,004
Divorces	110,160	1,724	3,733	30	35,338
Crop production value	90,720	1,645,574	2,194,976	13,650	19,592,152
Fertility rate	103,680	2.0	0.7	1.1	4.7
Hospitals	103,680	18.5	26.8	1	238
Students	84,240	66,985	115,474	2,695	990,111
Literacy rate (%)	110,160	95.0	3.3	78.7	99.1
Population	110,160	981,593	1,760,206	74,412	15,907,951
Population density	108,880	95.0	114.1	10.1	2,808
Students per classroom	84,240	21.0	5.6	12	48
Cereal production	110,160	1,367,386	1,749,240	335	19,253,855
Number of wildfires	77,760	34	50	0	322
Wildfire-affected area (hectares)	77,760	281	2,589	0	60,367
Foreign residents	110,160	7,922	43,100	24	675,678
Syrian refugees	90,720	32,730	86,826	0	655,505
Destination characteristics					
Average temperature (°C)	110,160	14.6	3.3	4.1	22.1
Mean maximum temperature (°C)	110,160	20.2	3.0	10.5	28.1
Mean minimum temperature (°C)	110,160	9.1	3.9	−2.8	18.6
Winter minimum temperature (°C)	110,160	0.6	5.0	−20.5	10.9
Summer maximum temperature (°C)	110,160	31.1	3.4	21.1	40.1
Total precipitation (mm)	110,160	643	322	121	3,097
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GDP (thousand TL)	110,160	91,518,579	486,800,000	568,507	13,000,000,000
GDP per capita (USD)	110,160	8,195	3,498	2,713	24,452
GDP per capita (TL)	110,160	65,239	104,615	3,888	802,669

Note: The table reports summary statistics for all variables used in the analysis. Most variables are observed for the full sample period, resulting in 110,160 observations. However, several origin-level socioeconomic variables have fewer observations due to limited temporal coverage. In particular, data on Syrian refugees and crop production values are available for a shorter time span, while information on fertility rates, hospitals, students, and students per classroom is reported only for selected years. These differences in data availability explain the variation in the number of observations across variables. *Data Source:* Turkish Statistical Institute (TurkStat) Data Source: TSI, TSMS

Distance data were taken from the Republic of Türkiye's General Directorate of Highways.[17] Distances are measured along the road network rather than as straight-line (Euclidean) distances and are treated as time-invariant. Distances between provinces in the dataset range from about 30 km to just over 2,000 km.

Population data for origin provinces were collected from TSI.[18] Within the sample, provincial populations range from approximately 75,000 in the smallest provinces to nearly 16 million in Istanbul, the country's most populous city.

Economic variables were obtained from the Turkish Statistical Institute (TSI). Provincial GDP per capita, measured in U.S. dollars, ranges from approximately \$2,700 to over \$24,000 across provinces and years. Total provincial GDP is reported in Turkish lira. Electricity consumption is measured in kilowatt-hours per capita and varies from fewer than 500 to over 11,000 kWh. Additional socioeconomic indicators include agricultural production values, fertility rates, hospitals, students, literacy rates, population levels, and population density, all reported at the provincial level by TSI.

Environmental stress indicators include wildfire data obtained from the General Directorate of Forestry in Türkiye.[19] These data record both the annual number of wildfire incidents and the total wildfire-affected area, measured in hectares. Across provinces and years, the number of wildfires ranges from 0 to over 300, while the affected area ranges from 0 to approximately 60,000 hectares.

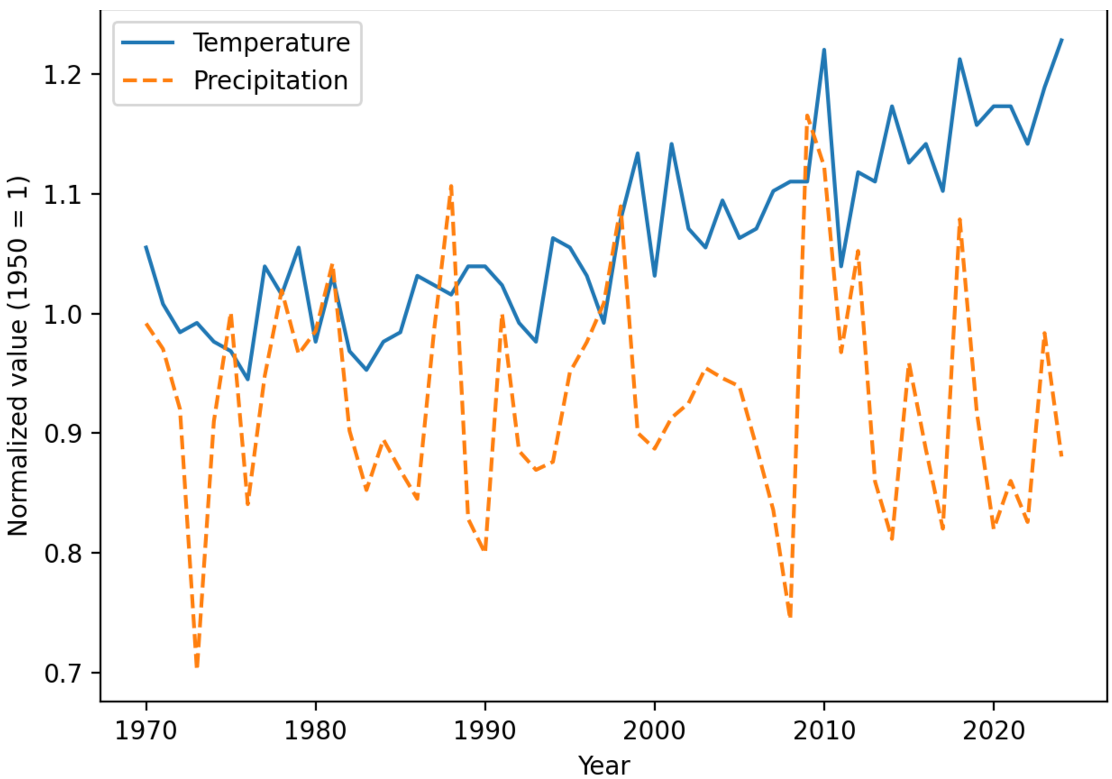

Figure 5: Normalized Temperature and Precipitation in Türkiye, 1970–2024

Note: Temperature and precipitation series are normalized relative to their 1950 values (1950 = 1) to facilitate comparison across variables measured in different units. The choice of 1950 as the reference year reflects a common baseline in climate studies and precedes the period of pronounced warming observed in the later decades.

Information on refugees was obtained from the Türkiye Presidency of Migration Management.[20] Provincial counts of Syrian refugees are included to capture potential migration pressures associated with large-scale

[17]Data Source: https://www.kgm.gov.tr/
[18]Data Source: https://data.tuik.gov.tr
[19]Data source: https://www.ogm.gov.tr/
[20]Data source: https://en.goc.gov.tr/

refugee inflows following the Syrian conflict. In addition, we construct a binary indicator that takes the value of one from 2014 onward, corresponding to the establishment of the Directorate General of Migration Management (DGMM).[21] This indicator captures institutional changes in the management of refugee inflows over time.

Figure 5 illustrates trends in average temperature and precipitation in Türkiye over the period 1970-2024, with both series normalized to their 1950 values. The figure shows a clear increase in average temperatures over time, accompanied by a gradual decline in precipitation. These patterns are consistent with findings in the literature (Beine and Parsons, 2017; Berlemann and Steinhardt, 2017). In addition, temperature and precipitation series exhibit increasing variability, indicating more pronounced climatic fluctuations in recent decades. Longer-run climate trends are reported in Appendix A, Figures A1 and A2.

6 Identification Strategy

Our identification strategy exploits variation in climatic conditions across provinces and over time to estimate the relationship between environmental factors and internal migration flows in Türkiye. The key explanatory variables are seasonal temperature and precipitation measures, including summer maximum temperatures and winter minimum temperatures, which capture periods of heightened climatic stress.

The empirical specification includes origin, destination, and year fixed effects, which absorb all time-invariant differences across provinces and common macroeconomic shocks. As a result, identification is based on within-province variation in climatic conditions over time. This approach isolates the effect of short-run fluctuations in environmental conditions from persistent geographic and climatic differences across regions.

To capture the potential push-and-pull mechanisms of climate on migration, we include temperature and precipitation variables for both origin and destination provinces. Adverse climatic conditions in origin provinces may increase out-migration by reducing agricultural productivity, worsening living conditions, or increasing environmental risks. Conversely, more favorable climatic conditions in destination provinces may attract migrants seeking better environmental and economic conditions.

To isolate the impact of climatic variables, we include several control variables that capture economic and spatial determinants of migration. GDP per capita at both origin and destination provinces controls for differences in economic opportunities that may influence migration decisions. Geographic distance between provinces proxies migration costs and spatial frictions, consistent with the gravity framework. In addition, we include variables capturing social pressures and environmental shocks, such as wildfire exposure, electricity consumption as a proxy for regional infrastructure and adaptation capacity, and the share of Syrian refugees in the origin province.

The empirical specification further incorporates origin, destination, and year fixed effects to control for unobserved heterogeneity. Origin fixed effects capture time-invariant characteristics of provinces such as geography, long-term economic structure, and cultural factors. Destination fixed effects control for persistent differences in destination attractiveness. Year-fixed effects account for national shocks that may simultaneously affect migration patterns across provinces. By absorbing these sources of unobserved heterogeneity, the identification relies on within-province variation in climatic conditions over time.

For estimation, we employ three complementary methods. First, we estimate the model using a three-way fixed effects specification to control for unobserved spatial and temporal heterogeneity. Second, we apply the PPML estimator, which has become the standard approach for gravity-type models because it accommodates zero migration flows and provides consistent estimates in the presence of heteroskedasticity (Silva and Tenreyro, 2006a). Finally, we implement the Hausman-Taylor estimator to address potential endogeneity while allowing the estimation of time-invariant variables such as geographic distance (Hausman and Taylor, 1981; Wooldridge, 2010).

Together, these approaches allow us to identify the relationship between climatic conditions and internal migration while controlling for economic opportunities, spatial frictions, and demographic pressures that may also influence migration decisions.

[21] The DGMM was established in 2014 to oversee migration and refugee-related policies in Türkiye.

7 Empirical Results

7.1 Graphical Analysis of Extreme Temperatures and Migration

This subsection presents graphical patterns between extreme temperatures and internal migration. We describe the figures sequentially to illustrate how migration flows vary with temperature conditions in origin and destination locations.

Figure 6 plots binned averages of migration flows against origin summer maximum temperature. Migration flows initially increase as summer temperatures rise, suggesting that higher temperatures in origin locations may act as a push factor. However, this pattern does not continue indefinitely. After a certain temperature level, the relationship begins to flatten and then slightly decline, indicating that the migration response to further increases in temperature weakens at very high temperatures. The quadratic fit highlights this nonlinear pattern.

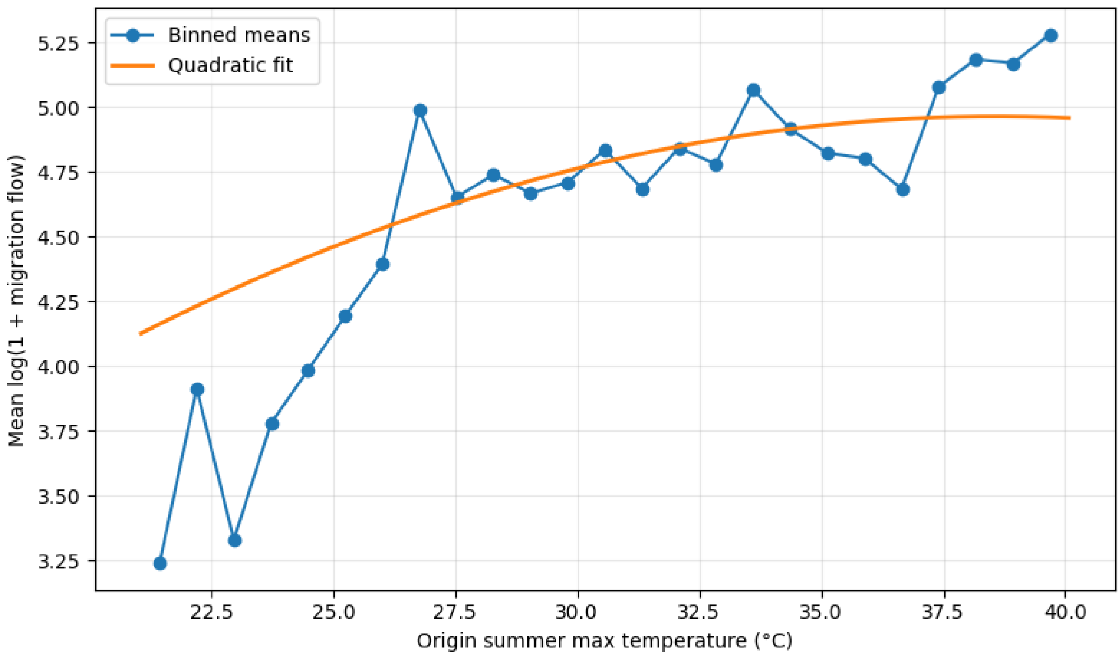

Figure 6: Migration Flows and Origin Summer Maximum Temperature

Note: The figure plots binned averages of log(1 + migration flows) against origin summer maximum temperature. The solid line represents a quadratic fit.

Figure 7 plots binned averages of migration flows against destination winter minimum temperature. Migration flows are relatively low toward destinations with very low temperatures and increase as winter minimum temperatures rise. This pattern indicates that migrants are more likely to move toward destinations with more moderate temperatures. The quadratic fit highlights a nonlinear relationship.

Additional graphical evidence is provided in Appendix B. Figures B1–B10 present alternative temperature–migration relationships using different temperature measures and normalization approaches. The qualitative patterns are consistent with those in the main text.

Figure 8 reports predicted migration rates across temperature categories, derived from a PPML gravity model that controls for distance, destination GDP, and population. These results reveal a distinct nonlinear "thermal optimum" in Turkish internal migration.Across all origin groups, migration probability peaks in the *Moderate* destination category. The drop in migration for *Cold* ($< 0°C$) and *Hot* ($> 32°C$) destinations indicates a clear "climate penalty," where extreme temperatures reduce a city's attractiveness. Furthermore, the figure highlights origin-specific push factors. The line for *Hot Origins* sits vertically higher than the others, meaning individuals in heat-stressed environments have a higher baseline probability of migrating

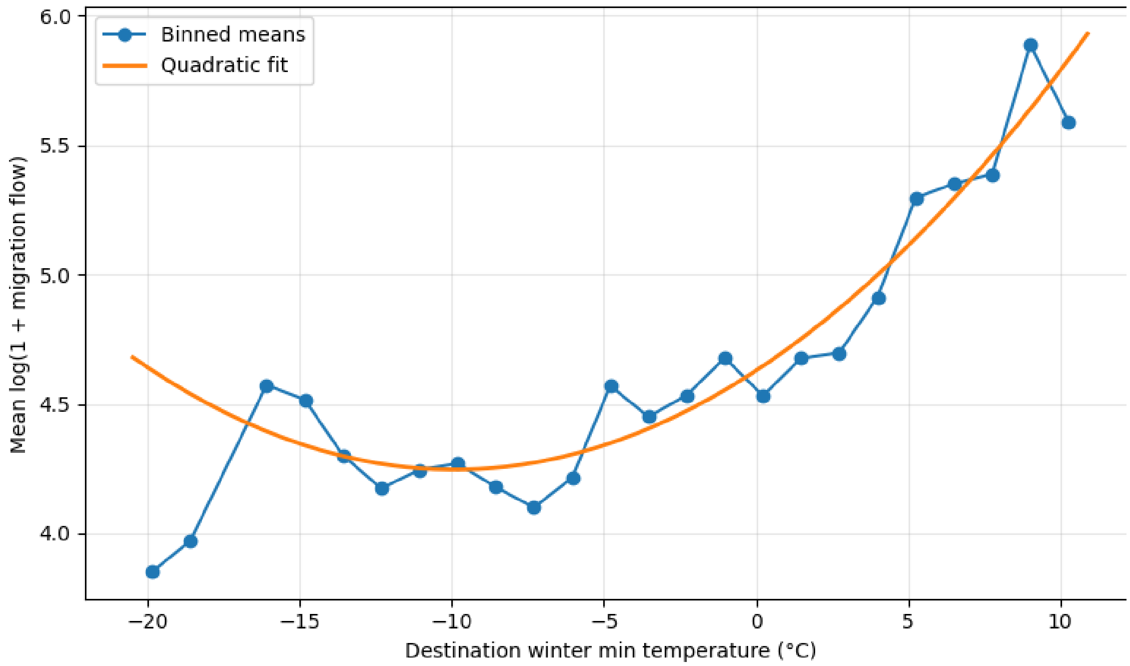

Figure 7: Migration Flows and Destination Winter Minimum Temperature

Note: The figure plots binned averages of $\log(1 + \text{migration flows})$ against destination winter minimum temperature. The solid line represents a quadratic fit.

regardless of where they are going. This confirms that extreme heat is a significant push factor, while moderate temperatures are the primary pull factor.

7.2 Full Sample Results

This subsection reports baseline regression results using the full sample of provinces in Türkiye. The baseline specification includes three-way fixed effects controlling for origin, destination, and year. The model is estimated using alternative estimation methods, including Poisson Pseudo-Maximum Likelihood (PPML) and the Hausman–Taylor estimator. These results provide an overview of the main determinants of internal migration and serve as a benchmark for the subsample analyses presented later.

7.2.1 Baseline Three-Way Fixed Effects Model

Table 2 reports the baseline regression results for the full sample of 81 provinces covering the period 2008–2024. The models are estimated using a stepwise specification to examine the effects of climatic conditions and environmental shocks on internal migration flows. All specifications include three-way fixed effects controlling for origin, destination, and year.

The results indicate a nonlinear relationship between temperature and migration. While the linear temperature coefficient becomes small in the full specifications, the quadratic temperature term remains positive and statistically significant ($\beta = 109.548$, $p < 0.01$). This pattern suggests that migration responses increase as temperatures move away from moderate levels, with higher temperatures generating stronger migration pressures.

Precipitation also exhibits a nonlinear relationship with migration. The linear precipitation coefficient is negative and statistically significant ($\beta = -0.050$, $p < 0.01$), while the squared term is positive and significant ($\beta = 0.012$, $p < 0.05$). This pattern suggests that migration flows increase when precipitation deviates from moderate levels, particularly during periods of low rainfall.

Environmental shocks also influence migration decisions. Beginning in Model (3), wildfire activity is included as an additional environmental stress indicator. The wildfire variable is positive and statistically

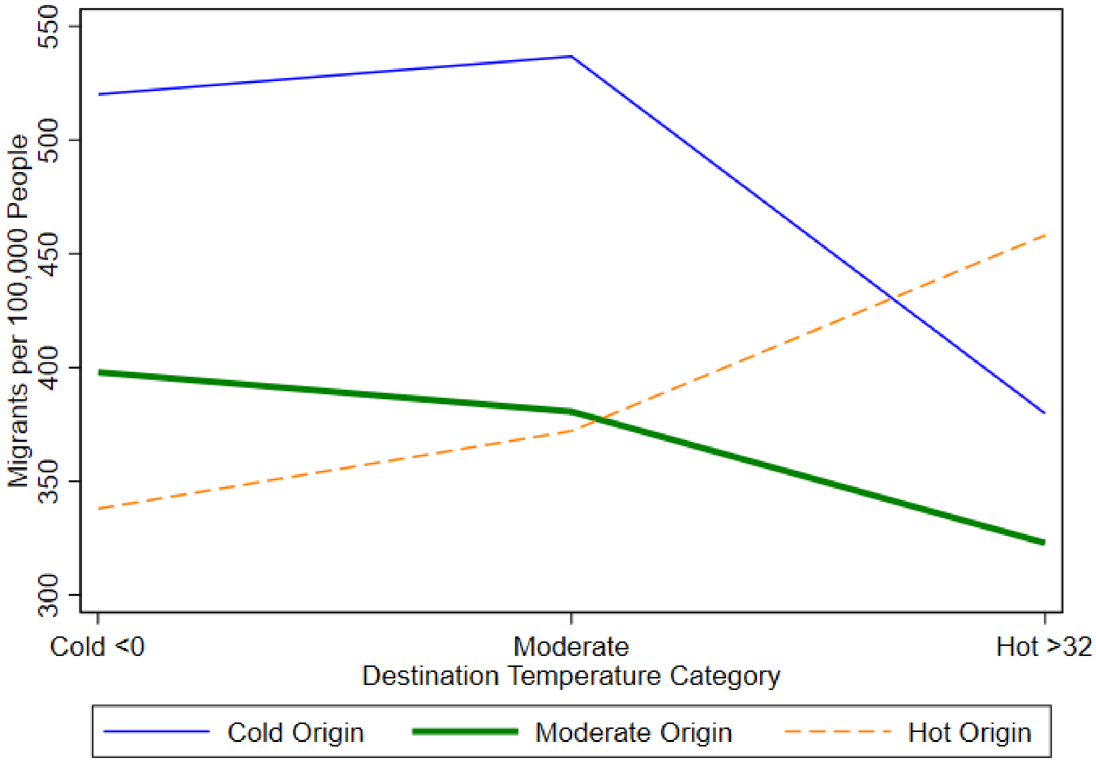

Figure 8: Predicted Migration Rates by Origin and Destination Temperature Categories

Note: The figure reports predicted migration rates (migrants per 100,000 people) across destination temperature categories.Estimates are obtained from a gravity model estimated using PPML, controlling for distance, destination GDP per capita, and destination population. Temperature categories are defined as Cold (<0°C), Moderate, and Hot (>32°C). Lines correspond to migrants from cold, moderate, and hot origin regions.

significant across specifications, indicating that regions experiencing larger wildfire events tend to generate higher migration flows.

Model (4) introduces electricity consumption as a proxy for economic activity and infrastructure development. The coefficient is positive and statistically significant, suggesting that regions with higher levels of economic activity are associated with greater migration flows, possibly reflecting improved economic capacity and mobility.

Finally, Model (6) incorporates the share of Syrian refugees in the origin province. The refugee variable is not statistically significant, suggesting that refugee presence does not systematically influence internal migration patterns at the national level.

Across all specifications, the coefficients on the climate variables remain stable, suggesting that the relationship between temperature, precipitation, and migration is robust to the inclusion of environmental shocks and socioeconomic controls.

Table 2: Baseline Three-Way Fixed Effects Estimates

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	3.983***	5.074***	0.860	0.375	0.391	0.391
	(0.84)	(0.89)	(0.99)	(0.98)	(0.98)	(0.98)
Origin Precipitation (mm)	-0.015***	-0.009*	-0.046***	-0.050***	-0.050***	-0.050***
	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Origin Temperature^2		68.161***	115.304***	109.664***	109.548***	109.548***
		(22.23)	(24.13)	(23.99)	(23.99)	(23.99)
Origin Precipitation^2		0.026***	0.016***	0.012**	0.012**	0.012**
		(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Wildfire Area (ha)			0.008***	0.008***	0.009***	0.009***
			(0.00)	(0.00)	(0.00)	(0.00)
Electricity Consumption (kWh)				0.060***	0.061***	0.061***
				(0.01)	(0.01)	(0.01)
Refugee Share (%)					0.006	0.006
					(0.02)	(0.02)
Constant	4.778***	4.764***	4.824***	4.355***	4.342***	4.342***
	(0.01)	(0.01)	(0.01)	(0.10)	(0.12)	(0.12)
Observations	110,160	110,160	110,160	77,760	77,760	77,760
R-squared	0.741	0.741	0.850	0.862	0.862	0.862
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	No	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. Temperature is measured in Kelvin (K) and precipitation in millimeters (mm). Temperature is centered around its sample mean prior to estimation in order to reduce multicollinearity in the nonlinear specification. Climate and economic variables enter the regressions in logarithmic form unless otherwise indicated. Squared terms capture potential nonlinear climate effects. The models follow a step-wise progression to evaluate the contribution of climate variables and environmental shocks to internal migration across all 81 provinces of Türkiye. Spatial distance is not included in this specification because time-invariant spatial frictions are absorbed by the high-dimensional fixed effects. Model (6) presents the full specification including the refugee share variable. All regressions include three-way fixed effects (origin, destination, and year) to control for unobserved heterogeneity and nationwide temporal shocks. Robust standard errors clustered at the city-pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Overall, the model explains a substantial share of the variation in migration flows ($R^2 = 0.742$). The stability of the climate coefficients across specifications indicates that the relationship between temperature, precipitation, and migration remains robust after controlling for environmental shocks and socioeconomic factors.

7.2.2 PPML Estimates

Table 3 reports the PPML estimates for internal migration flows across the full sample of 81 provinces in Türkiye. The PPML estimator is commonly used to estimate gravity models because it accommodates zero flows and is robust to heteroskedasticity. The specifications follow the same stepwise structure used in the fixed-effects models, gradually introducing economic, geographic, and social controls.

The results provide strong evidence of nonlinear climate effects. Across all specifications, the squared temperature term for origin provinces remains positive and highly statistically significant. This indicates that migration responses intensify as temperatures rise, suggesting that extreme heat in origin locations acts as a push factor for internal migration.

In contrast, the squared temperature term for destination provinces is not statistically significant across specifications. This asymmetry suggests that while high temperatures in origin regions encourage residents to leave, migrants do not appear to select destination provinces based on a similar nonlinear temperature pattern.

Precipitation also plays a role in migration dynamics. The coefficient on origin precipitation is generally negative, indicating that lower rainfall levels are associated with higher migration flows. Destination precipitation also enters with a negative coefficient, suggesting that climatic conditions at both origin and destination locations influence migration decisions.

Economic factors further shape migration flows. Higher GDP per capita in destination provinces is associated with greater migration inflows, suggesting that migrants are attracted to more economically advanced regions. In contrast, the coefficient on origin GDP per capita is negative, suggesting that residents in less prosperous provinces are more likely to migrate.

Geographic distance enters the regression in Models (5) and (6) with the expected negative sign. The magnitude of the coefficient indicates that migration flows decline substantially with increasing distance between provinces, consistent with standard gravity model predictions.

Finally, Model (6) introduces the share of Syrian refugees in the origin province. The coefficient is positive and statistically significant, suggesting that higher refugee presence is associated with increased internal migration flows among the domestic population.

Overall, the PPML estimates reinforce the findings from the fixed-effects models. Climatic conditions—particularly rising temperatures in origin provinces—play an important role in shaping internal migration patterns in Türkiye. The final specification also exhibits a strong model fit, with a pseudo-R^2 of 0.863.

Table 3: PPML Gravity Model Estimates for Internal Migration: Full Sample of 81 Provinces in Türkiye

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	4.277**	4.776**	5.034***	5.078***	4.611**	1.661
	(1.941)	(1.924)	(1.937)	(1.932)	(1.921)	(2.284)
Origin Temperature^2	178.338***	157.569***	158.847***	159.333***	163.866***	141.861***
	(45.158)	(45.093)	(44.615)	(44.325)	(44.111)	(45.507)
Origin Precipitation (mm)	-0.020*	-0.018	-0.018	-0.018	-0.016	-0.027**
	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)	(0.013)
Origin GDP per capita (USD)		-0.229***	-0.228***	-0.226***	-0.233***	-0.078*
		(0.054)	(0.054)	(0.053)	(0.054)	(0.046)
Destination Temperature (K)			6.783***	6.449***	5.928***	6.102***
			(1.599)	(1.618)	(1.599)	(1.905)
Destination Temperature^2			-45.351	-31.174	-48.033	-13.919
			(33.182)	(33.535)	(33.610)	(38.613)
Destination Precipitation (mm)			-0.019**	-0.021**	-0.019*	-0.031**
			(0.010)	(0.010)	(0.010)	(0.012)
Destination GDP per capita (USD)				0.164***	0.177***	0.187***
				(0.046)	(0.046)	(0.047)
Distance (km)					-0.764***	-0.768***
					(0.024)	(0.023)
Refugee Share (%)						0.025***
						(0.009)
Constant	7.002***	9.100***	9.197***	7.676***	12.409***	10.964***
	(0.022)	(0.491)	(0.487)	(0.642)	(0.628)	(0.644)
Observations	110,160	110,160	110,160	110,160	110,160	90,720
Pseudo R^2	0.762	0.762	0.762	0.762	0.857	0.863
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is migration flows between cities and is estimated in levels using the Poisson pseudo–maximum likelihood (PPML) estimator. Independent variables enter the regressions in logarithmic form unless otherwise indicated. Temperature variables represent the average temperature measured in Kelvin (K), while precipitation is measured in millimeters (mm). Climate variables include both origin and destination temperature and precipitation conditions. All specifications include high-dimensional fixed effects for *Origin*, *Destination*, and *Year* to account for time-invariant spatial heterogeneity and common temporal shocks. Standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

7.2.3 Hausman-Taylor Estimation

Table 4 presents the Hausman-Taylor estimates for internal migration flows across the full sample of 81 provinces in Türkiye for the period 2008–2024. The Hausman-Taylor estimator allows the inclusion of time-invariant variables, such as geographic distance, while addressing potential endogeneity in the selected regressors. In this specification, origin GDP per capita is treated as endogenous, whereas climate variables and refugee shares are treated as exogenous.

The results again provide strong evidence of nonlinear climate effects. The squared temperature term remains positive and highly statistically significant across specifications. In Model (6), the coefficient of the squared temperature term equals 149.3 ($p < 0.01$), indicating that migration responses intensify as temperatures rise beyond moderate levels. This finding is consistent with the nonlinear temperature effects observed in the fixed effects and PPML gravity estimations.

Precipitation also exhibits nonlinear effects. The linear precipitation coefficient is negative and statistically significant, while the squared term is positive and significant across specifications. In Model (6), the coefficient on precipitation equals -0.230 ($p < 0.01$), while the squared term equals 0.0146 ($p < 0.01$). These estimates suggest that lower precipitation levels are associated with increased out-migration from origin provinces.

Environmental shocks further influence migration flows. Beginning in Model (4), wildfire activity enters the regression with a positive and statistically significant coefficient. In the full specification, the wildfire coefficient equals 0.00545, indicating that provinces experiencing larger wildfire events tend to generate higher migration flows.

Economic variables also play an important role. Origin GDP per capita, treated as endogenous in the Hausman-Taylor framework, remains negative and statistically significant across specifications. This result suggests that weaker economic conditions in origin provinces are associated with stronger out-migration pressures. Electricity consumption, introduced in Model (6), is positive and statistically significant, indicating that higher economic activity is associated with greater migration flows.

Geographic distance remains a key determinant of migration flows. The coefficient on log distance is consistently negative and statistically significant across all specifications, with an estimated magnitude of approximately -0.70. This finding is consistent with gravity-type migration models in which spatial frictions reduce migration flows between distant provinces.

Finally, the refugee share variable in Models (5) and (6) is positive and statistically significant, indicating that greater refugee presence is associated with higher internal migration flows.

Overall, the Hausman-Taylor estimates reinforce the main findings of the previous specifications. Nonlinear temperature effects, precipitation pressures, and environmental shocks continue to play an important role in shaping internal migration patterns across Türkiye, while geographic distance remains a significant constraint on mobility.

Table 4 reports the Hausman–Taylor estimates for internal migration flows across the 81 provinces of Türkiye over the period 2008–2024. The Hausman–Taylor estimator allows the inclusion of time-invariant variables, such as geographic distance, while addressing potential endogeneity of economic variables. In this specification, log origin GDP per capita is treated as endogenous, while climate variables and refugee shares are treated as exogenous shocks.

The results confirm the presence of non-linear climate effects. The coefficient on the squared temperature term remains positive and statistically significant across all specifications. In Model (6), the coefficient of the squared temperature term is 149.3 ($p < 0.01$), indicating that migration responses increase as temperature rises beyond moderate levels. This pattern is consistent with the evidence obtained from the fixed effects and PPML estimations.

Precipitation effects are also non-linear. The linear precipitation coefficient is negative and statistically significant, while the squared precipitation term is positive and significant. In Model (6), the coefficient for log precipitation is -0.230 ($p < 0.01$), and the squared term equals 0.0146 ($p < 0.01$). These estimates suggest that lower precipitation levels are associated with higher out-migration, consistent with drought-related pressures.

Environmental shocks also influence migration flows. The wildfire variable becomes positive and significant when introduced in Model (4), with a coefficient of 0.00545 in the full specification. Similarly, the refugee share variable introduced in Model (5) shows a positive and statistically significant effect (0.138,

$p < 0.01$), indicating that higher refugee presence is associated with increased internal mobility.

Economic and geographic variables behave as expected. Log distance is consistently negative across all models, with a coefficient of approximately -0.70, confirming the presence of spatial frictions in internal migration. Origin GDP per capita, treated as endogenous, remains negative and statistically significant in all specifications, suggesting that lower economic conditions in origin provinces are associated with stronger migration outflows.

Overall, the Hausman–Taylor estimates support the main findings of the previous specifications. Nonlinear temperature effects, drought pressures, and environmental shocks contribute to internal migration patterns across Türkiye, while distance continues to represent a significant constraint on mobility.

Table 4: Hausman-Taylor Estimates for Internal Migration: Full Sample of 81 Provinces in Türkiye

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	7.877***	9.369***	9.192***	6.967***	4.769***	3.963***
	(0.579)	(0.615)	(0.623)	(0.721)	(0.736)	(0.743)
Origin Temperature^2		103.6***	102.0***	179.9***	158.5***	149.3***
		(14.96)	(15.01)	(19.88)	(19.90)	(19.92)
Origin Precipitation (mm)			-0.314***	-0.245***	-0.280***	-0.230***
			(0.0656)	(0.0702)	(0.0701)	(0.0704)
Origin Precipitation^2			0.0245***	0.0166***	0.0189***	0.0146***
			(0.00519)	(0.00560)	(0.00559)	(0.00562)
Wildfire Area (ha)				0.00424***	0.00502***	0.00545***
				(0.000871)	(0.000871)	(0.000872)
Origin GDP per capita (USD)	-0.120***	-0.121***	-0.120***	-0.0757***	-0.0608***	-0.111***
	(0.0139)	(0.0140)	(0.0140)	(0.0157)	(0.0157)	(0.0161)
Refugee Share (%)					0.145***	0.138***
					(0.00994)	(0.00997)
Electricity Consumption (kWh)						0.0719***
						(0.00950)
Distance (km)	-0.709***	-0.712***	-0.712***	-0.727***	-0.704***	-0.702***
	(0.0253)	(0.0252)	(0.0253)	(0.0249)	(0.0251)	(0.0251)
Constant	10.19***	10.20***	11.19***	10.85***	10.68***	10.09***
	(0.208)	(0.208)	(0.292)	(0.306)	(0.308)	(0.311)
Observations	110,160	110,160	110,160	77,760	77,760	77,760
Pair Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. Estimations are conducted using the Hausman–Taylor (1981) instrumental variables estimator, which allows the inclusion of time-invariant variables while addressing potential endogeneity of selected regressors. Temperature variables represent average temperature measured in Kelvin (K), while precipitation is measured in millimeters (mm). Squared terms capture potential nonlinear climate effects. Economic and environmental controls include GDP per capita (USD), wildfire area (hectares), electricity consumption (kWh), refugee share (%), and geographic distance between provinces (km). All explanatory variables enter the regressions in logarithmic form unless otherwise indicated. Origin GDP per capita is treated as an endogenous variable, while climate variables and refugee shares are treated as exogenous. Geographic distance is identified through the internal instrumentation procedure of the Hausman-Taylor estimator. All specifications include origin–destination pair fixed effects and year fixed effects. Standard errors clustered at the city-pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

7.3 Climate-Exposed Provinces: 30-City Subsample

While the full-sample analysis provides a national overview of migration dynamics, this subsection focuses on a subsample of 30 provinces identified as highly exposed to climatic stress. Restricting the analysis to these provinces allows us to examine whether migration responses to climatic conditions are stronger in environmentally vulnerable regions. The 30 provinces included in this subsample are selected based on their exposure to extreme temperatures and climatic variability.

The empirical strategy follows the same approach used in the national analysis. Specifically, we estimate three sets of models: a baseline specification with three-way fixed effects, a PPML gravity model, and a

Hausman-Taylor estimator that accounts for potential endogeneity while allowing the inclusion of time-invariant variables such as geographic distance.

7.3.1 Three-Way Fixed Effects Results: 30-City Subsample

Table 5 reports the estimation results for the subsample of 30 climate-exposed provinces using a three-way fixed effects specification. As in the national analysis, the models include origin, destination, and year fixed effects to control for time-invariant heterogeneity and common temporal shocks.

The results indicate that temperature effects are larger in magnitude compared with the national sample. In the full specification (Model 6), the coefficient on origin temperature equals $\beta = 13.762$ ($p < 0.01$), suggesting that higher temperatures in origin provinces are strongly associated with increased migration flows. This finding indicates that migration responses to rising temperatures are more pronounced in provinces with high exposure to climatic stress.

In contrast, the squared temperature term is not statistically significant across specifications. This suggests that, within climate-exposed regions, migration responses increase with temperature but do not display a clearly nonlinear pattern.

Precipitation continues to exhibit a negative and statistically significant relationship with migration flows. In Model (6), the coefficient on origin precipitation equals $\beta = -0.107$ ($p < 0.01$), indicating that lower rainfall levels are associated with higher migration outflows. The squared precipitation term becomes positive and statistically significant in later specifications, suggesting that migration responses intensify as precipitation levels decline further.

Table 5: Three-Way Fixed Effects Estimates: Climate-Exposed Subsample of 30 Origin Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	10.421***	10.965***	13.724***	13.322***	13.762***	13.762***
	(1.76)	(1.72)	(1.82)	(1.80)	(1.81)	(1.81)
Origin Precipitation (mm)	-0.054***	-0.051***	-0.105***	-0.108***	-0.107***	-0.107***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Origin Temperature^2		50.405	69.836	63.011	56.810	56.810
		(52.92)	(57.56)	(56.62)	(56.90)	(56.90)
Origin Precipitation^2		0.007	0.015*	0.016**	0.017**	0.017**
		(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Wildfire Area (ha)			0.010***	0.011***	0.011***	0.011***
			(0.00)	(0.00)	(0.00)	(0.00)
Electricity Consumption (kWh)				0.021	0.022	0.022
				(0.02)	(0.02)	(0.02)
Refugee Share (%)					0.036	0.036
					(0.04)	(0.04)
Constant	4.678***	4.669***	4.731***	4.576***	4.477***	4.477***
	(0.01)	(0.01)	(0.02)	(0.14)	(0.17)	(0.17)
Observations	40,800	40,800	40,800	28,800	28,800	28,800
R-squared	0.725	0.725	0.853	0.866	0.866	0.866
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	No	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. Temperature is measured in Kelvin (K) and precipitation in millimeters (mm). Wildfire area is measured in hectares (ha), electricity consumption in kilowatt-hours (kWh), and the refugee variable represents the percentage share of refugees in the origin province. The sample is restricted to 30 origin provinces identified as highly exposed to climatic stress, while all 81 provinces are included as potential destinations. The models follow a stepwise specification to evaluate climate effects and additional environmental and socioeconomic controls. All regressions include origin, destination, and year fixed effects. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Environmental shocks also contribute to migration dynamics. The wildfire variable becomes positive and statistically significant once introduced in Model (3), with a coefficient of $\beta = 0.011$ ($p < 0.01$) in the full specification. This result indicates that provinces experiencing larger wildfire events tend to generate higher migration flows.

Economic controls appear less influential in this subsample. Electricity consumption does not exhibit a statistically significant effect on migration flows. Similarly, the Syrian refugee share included in Models (5) and (6) is not statistically significant, suggesting that climatic and environmental pressures play a more prominent role in shaping migration outcomes in climate-exposed provinces.

Overall, the model explains a substantial share of migration variation within this subsample ($R^2 = 0.866$), indicating that climatic factors and environmental shocks are closely associated with migration dynamics in climate-vulnerable regions.

7.3.2 PPML Estimates: 30-City Subsample

Table 6 reports the PPML gravity model estimates for the subsample of 30 climate-exposed provinces. The specifications follow the same stepwise structure used in the national analysis and include origin, destination, and year fixed effects. Restricting the sample to climate-exposed provinces allows us to evaluate whether the climate–migration relationship differs in regions that face stronger environmental pressures.

Table 6: PPML Gravity Model Estimates: Climate-Exposed Subsample of 30 Origin Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	9.555***	11.117***	11.261***	11.239***	10.509***	8.479**
	(3.626)	(3.586)	(3.580)	(3.572)	(3.528)	(3.689)
Origin Temperature^2	175.401*	155.855	155.097	154.959	173.679*	277.373**
	(102.208)	(102.780)	(101.826)	(101.293)	(99.725)	(114.127)
Origin Precipitation (mm)	-0.122***	-0.115***	-0.115***	-0.115***	-0.112***	-0.157***
	(0.026)	(0.025)	(0.025)	(0.025)	(0.025)	(0.031)
Origin GDP per capita (USD)		-0.262***	-0.262***	-0.258***	-0.266***	-0.092
		(0.080)	(0.079)	(0.078)	(0.078)	(0.061)
Destination Temperature (K)			12.147***	11.262***	9.987***	10.080***
			(2.903)	(2.814)	(2.729)	(3.689)
Destination Temperature^2			-112.859	-93.875	-109.709	-41.168
			(69.244)	(69.734)	(68.629)	(74.781)
Destination Precipitation (mm)			-0.011	-0.015	-0.002	-0.016
			(0.018)	(0.018)	(0.018)	(0.023)
Destination GDP per capita (USD)				0.269***	0.285***	0.218***
				(0.066)	(0.067)	(0.072)
Distance (km)					-1.141***	-1.129***
					(0.041)	(0.040)
Refugee Share (%)						0.029**
						(0.015)
Constant	6.606***	8.900***	8.923***	6.441***	13.593***	12.643***
	(0.036)	(0.695)	(0.681)	(0.902)	(0.884)	(0.929)
Observations	40,800	40,800	40,800	40,800	40,800	33,600
Pseudo R-squared	0.736	0.736	0.736	0.737	0.873	0.876
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is migration flows between origin and destination provinces, estimated in levels using the Poisson pseudo–maximum likelihood (PPML) estimator. Temperature variables are measured in Kelvin (K) and precipitation in millimeters (mm). Distance is measured in kilometers (km), wildfire area in hectares (ha), electricity consumption in kilowatt-hours (kWh), and the refugee variable represents the percentage share of refugees in the origin province. The sample restricts origin provinces to the 30 cities identified as highly exposed to climatic stress, while all 81 provinces remain potential destinations. All specifications include origin, destination, and year fixed effects. Standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The results provide evidence of nonlinear temperature effects. Both the linear and squared temperature terms for origin provinces remain positive across specifications. In the full model (Model 6), the coefficient on origin temperature equals $\beta = 8.479$ ($p < 0.05$), while the squared temperature term equals $\beta = 277.373$ ($p < 0.05$). These estimates suggest that migration responses intensify as temperatures rise beyond moderate levels, indicating that extreme heat acts as a strong push factor in climate-exposed provinces.

Precipitation effects remain negative and statistically significant across all specifications. The coefficient on origin precipitation in Model (6) equals $\beta = -0.157$ ($p < 0.01$), indicating that lower rainfall levels are associated with higher migration outflows. This finding supports the interpretation that drought conditions contribute to migration pressures in environmentally vulnerable regions.

Economic variables also influence migration patterns. Destination GDP per capita is positively and statistically significant, suggesting that migrants are attracted to economically stronger provinces. In contrast, origin GDP per capita loses statistical significance in the full specification.

Geographic distance retains the expected negative sign, with a coefficient of approximately -1.129 in Model (6). The magnitude of this coefficient is larger than the corresponding estimate in the national sample, suggesting that spatial frictions play an important role in shaping migration flows among climate-exposed provinces.

Finally, the refugee share variable becomes positive and statistically significant in the full specification (0.029, $p < 0.05$), indicating that higher refugee presence is associated with increased internal migration flows.

Overall, the PPML estimates reinforce the findings from the fixed effects and Hausman–Taylor specifications. Climatic conditions—particularly extreme temperatures and reduced precipitation—continue to play an important role in shaping migration patterns within climate-exposed provinces. The final specification exhibits a strong model fit with a pseudo-R^2 of 0.876.

7.3.3 Hausman-Taylor Estimation: 30-City Subsample

Table 7 reports the Hausman-Taylor estimates for the subsample of 30 climate-exposed provinces. The Hausman–Taylor estimator allows the inclusion of time-invariant variables, such as geographic distance, while addressing potential endogeneity of selected regressors. In this specification, origin GDP per capita is treated as endogenous, whereas climate variables and refugee shares are treated as exogenous.

The results again provide strong evidence of temperature effects. Both the linear and squared temperature terms remain positive and statistically significant across specifications. In the full model (Model 6), the coefficient on origin temperature equals $\beta = 19.49$ ($p < 0.01$), while the squared temperature term equals $\beta = 131.3$ ($p < 0.01$). These estimates indicate that migration responses increase as temperatures rise and intensify further at higher temperature levels. This finding is consistent with the nonlinear temperature effects observed in the previous specifications.

Precipitation variables are not statistically significant in this subsample. This suggests that rainfall variations may play a smaller role in explaining migration flows among climate-exposed provinces compared with temperature-related pressures.

Environmental shocks continue to influence migration flows. Wildfire activity enters the regression in Model (4) with a positive and statistically significant coefficient. In the full specification, the wildfire coefficient equals 0.00865 ($p < 0.01$), indicating that provinces experiencing larger wildfire events tend to generate higher outflows of migration.

Economic and geographic factors also affect migration dynamics. Origin GDP per capita, treated as endogenous in the Hausman–Taylor framework, remains negative and statistically significant across specifications. In Model (6), the coefficient equals -0.110 ($p < 0.01$), suggesting that weaker economic conditions in origin provinces are associated with stronger migration pressures.

Geographic distance remains a key determinant of migration flows. The coefficient on distance is consistently negative and statistically significant across all specifications, with an estimated magnitude of approximately -0.615 in the full model. This result confirms that spatial frictions continue to constrain migration flows even among climate-exposed regions.

Finally, the refugee share does not exhibit a statistically significant effect in this subsample. Electricity consumption enters the full specification with a small positive coefficient and weak statistical significance ($p < 0.1$), suggesting a modest association between economic activity and migration flows.

Table 7: Hausman–Taylor Estimates: Climate-Exposed Subsample of 30 Origin Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	16.73***	17.54***	16.84***	19.87***	19.83***	19.49***
	(1.088)	(1.114)	(1.125)	(1.368)	(1.437)	(1.454)
Origin Temperature^2		101.0***	83.55***	135.6***	135.0***	131.3***
		(30.44)	(30.68)	(43.94)	(44.39)	(44.53)
Origin Precipitation (mm)			-0.103	-0.0221	-0.0221	-0.0439
			(0.104)	(0.109)	(0.109)	(0.110)
Origin Precipitation^2			0.00521	0.00870	0.00870	0.0106
			(0.00826)	(0.00877)	(0.00877)	(0.00883)
Wildfire Area (ha)				0.00844***	0.00844***	0.00865***
				(0.00160)	(0.00160)	(0.00160)
Origin GDP per capita (USD)	-0.243***	-0.241***	-0.237***	-0.0720***	-0.0697***	-0.110***
	(0.0209)	(0.0209)	(0.0209)	(0.0239)	(0.0243)	(0.0249)
Refugee Share (%)					0.00125	-0.00391
					(0.0143)	(0.0143)
Electricity Consumption (kWh)						0.0253*
						(0.0137)
Distance (km)	-0.602***	-0.603***	-0.600***	-0.616***	-0.616***	-0.615***
	(0.0403)	(0.0402)	(0.0404)	(0.0393)	(0.0393)	(0.0394)
Constant	10.39***	10.36***	10.74***	9.553***	9.529***	9.772***
	(0.322)	(0.322)	(0.459)	(0.483)	(0.489)	(0.492)
Observations	40,800	40,800	40,800	28,800	28,800	28,800
Pair Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows originating from a subsample of 30 climate-exposed provinces. Estimations are conducted using the Hausman–Taylor (1981) instrumental variables estimator, which allows the inclusion of time-invariant variables while addressing potential endogeneity of selected regressors.Temperature variables are measured in Kelvin (K) and precipitation in millimeters (mm). Squared terms capture potential nonlinear climate effects. Origin GDP per capita is treated as endogenous, while climate variables and refugee shares are treated as exogenous. All specifications include origin–destination pair fixed effects and year fixed effects. Standard errors clustered at the city-pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Overall, the Hausman-Taylor estimates reinforce the results obtained from the fixed effects and PPML gravity models. Temperature increases and environmental shocks remain important factors shaping migration dynamics in climate-exposed provinces.

7.4 High-Exposure Provinces: 17-City Drought Subsample

To further examine the relationship between climate conditions and internal migration, this subsection focuses on a smaller group of 17 provinces identified as highly exposed to drought conditions. Restricting the analysis to these provinces allows us to investigate whether migration responses become stronger in regions that face the most severe climatic pressures.

The empirical strategy follows the same approach used in the previous sections. Specifically, we estimate three models: a baseline specification with three-way fixed effects, a PPML gravity model, and a Hausman-Taylor estimator that accounts for potential endogeneity while allowing the inclusion of time-invariant variables, such as geographic distance.

7.4.1 Three-Way Fixed Effects Results: Drought-Exposed Provinces

Table 8 reports the fixed effects estimates for the subsample of 17 provinces that exhibit the highest exposure to drought conditions and environmental stress. These provinces are primarily located in Türkiye's Southeastern and Mediterranean regions and represent areas where climatic pressures and socioeconomic vulnerabilities overlap.

The results indicate that precipitation plays a particularly important role in explaining migration dynamics within this subsample. The coefficient on origin precipitation remains negative and statistically significant across all specifications. In the full model (Model 6), the estimated coefficient equals $\beta = -0.113$ ($p < 0.01$), indicating that lower rainfall levels are associated with higher migration outflows. This finding highlights the importance of water scarcity in drought-prone provinces, where agricultural dependence and climatic variability strongly influence migration decisions.

Temperature effects also appear relevant in these high-exposure regions. While the linear temperature coefficient becomes statistically significant only in later specifications, the squared temperature term is significant in intermediate models, suggesting nonlinear temperature responses. However, the squared term loses statistical significance in the full specification, indicating that the temperature effect is less precisely estimated once additional controls are introduced.

Environmental shocks appear less influential in this subsample. The wildfire variable enters the regression in Model (3) but is not statistically significant in the subsequent specifications. This suggests that gradual climatic pressures, particularly drought conditions, may play a more consistent role than short-term environmental shocks in explaining migration flows in these provinces.

Economic and social factors also contribute to migration dynamics. Electricity consumption, used as a proxy for local economic capacity and infrastructure, becomes positive and statistically significant in the later specifications. In the full model, the coefficient is 0.059 ($p < 0.05$), suggesting that higher levels of development may facilitate households' ability to relocate when facing climatic or economic pressures.

A notable result in this subsample is the strong role of demographic pressures. The share of Syrian refugees becomes positive and highly statistically significant in Models (5) and (6), with an estimated coefficient of 0.486 ($p < 0.01$). This finding suggests that provinces located near the Syrian border experience additional migration pressures where climatic vulnerability and demographic shocks overlap.

Table 8: Three-Way Fixed Effects Estimates: High-Exposure Subsample of 17 Origin Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	0.572	1.477	1.973	1.000	6.616**	6.616**
	(2.27)	(2.18)	(2.61)	(2.60)	(2.57)	(2.57)
Origin Precipitation (mm)	-0.078***	-0.063***	-0.110***	-0.116***	-0.113***	-0.113***
	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)	(0.02)
Origin Temperature^2		255.341***	221.364**	207.128**	98.964	98.964
		(76.37)	(89.30)	(88.65)	(89.18)	(89.18)
Origin Precipitation^2		0.022**	0.023**	0.026**	0.028***	0.028***
		(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Wildfire Area (ha)			0.008	0.007	0.008	0.008
			(0.01)	(0.01)	(0.01)	(0.01)
Electricity Consumption (kWh)				0.039*	0.059**	0.059**
				(0.02)	(0.02)	(0.02)
Refugee Share (%)					0.486***	0.486***
					(0.05)	(0.05)
Constant	4.802***	4.758***	4.849***	4.567***	2.874***	2.874***
	(0.02)	(0.02)	(0.03)	(0.18)	(0.23)	(0.23)
Observations	23,120	23,120	23,120	16,320	16,320	16,320
R-squared	0.819	0.819	0.884	0.896	0.896	0.897
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. Temperature variables are measured in Kelvin (K) and precipitation in millimeters (mm). Wildfire area is measured in hectares (ha), electricity consumption in kilowatt-hours (kWh), and the refugee variable represents the percentage share of refugees in the origin province.The sample restricts origin provinces to 17 cities identified as highly exposed to drought conditions and refugee inflows, while all 81 provinces remain potential destinations. All regressions include origin, destination, and year fixed effects. Robust standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Overall, the results indicate that migration dynamics in these drought-prone provinces are strongly shaped by environmental pressures. In particular, the magnitude and consistency of the precipitation and refugee coefficients highlight the interaction between climatic vulnerability and social stress in driving migration flows within these high-exposure regions.

7.4.2 PPML Estimates: 17-City Drought Subsample

Table 9 reports the PPML estimates for migration flows originating from the 17 provinces most affected by drought conditions in Türkiye. The PPML estimator is well-suited for migration gravity models because it accommodates zero flows and provides consistent estimates in the presence of heteroskedasticity.

The results indicate that temperature remains a strong determinant of migration in drought-exposed regions. Across all specifications, the coefficient on origin temperature is positive and statistically significant. In the full specification (Model 6), the estimated coefficient equals $\beta = 31.440$ ($p < 0.01$), indicating that higher temperatures are associated with increased migration outflows from drought-affected provinces. In contrast, the squared temperature term is not statistically significant, suggesting that the relationship between temperature and migration is primarily linear within this highly exposed subsample.

Precipitation also contributes to migration pressures in drought-prone regions. While the precipitation coefficient is not always significant in the early specifications, it becomes negative and highly significant in the later models. In Model (6), the coefficient equals $\beta = -0.208$ ($p < 0.01$), indicating that lower rainfall levels are associated with higher migration outflows.

Table 9: PPML Gravity Model Estimates: Drought-Exposed Subsample of 17 Origin Provinces

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	49.225***	55.733***	54.443***	50.908***	44.082***	31.440***
	(2.736)	(3.165)	(3.174)	(2.901)	(3.001)	(4.696)
Origin Temperature^2	179.283	194.139	303.056	198.046	221.897	83.444
	(202.352)	(220.121)	(213.168)	(198.219)	(200.242)	(244.646)
Origin Precipitation (mm)	-0.027	-0.094***	-0.059*	-0.001	-0.155***	-0.208***
	(0.035)	(0.035)	(0.034)	(0.034)	(0.033)	(0.038)
Origin GDP per capita (USD)		0.415***	0.372***	0.151***	0.304***	0.239***
		(0.041)	(0.040)	(0.041)	(0.046)	(0.049)
Destination Temperature (K)			47.127***	44.720***	33.676***	32.771***
			(0.865)	(0.985)	(0.950)	(1.001)
Destination Temperature^2			-236.137***	-661.092***	-481.836***	-469.911***
			(66.287)	(55.161)	(56.049)	(60.428)
Destination Precipitation (mm)			-0.126***	-0.247***	-0.031	-0.036
			(0.032)	(0.032)	(0.031)	(0.033)
Destination GDP per capita (USD)				0.991***	2.009***	1.930***
				(0.055)	(0.061)	(0.067)
Distance (km)					-1.009***	-0.983***
					(0.019)	(0.021)
Refugee Share (%)						0.052***
						(0.010)
Constant	5.404***	1.780***	2.826***	-3.476***	-3.530***	-3.927***
	(0.023)	(0.360)	(0.389)	(0.608)	(0.533)	(0.591)
Observations	23,120	23,120	23,120	23,120	23,120	19,040
Pseudo R-squared	0.076	0.087	0.205	0.264	0.423	0.426
Origin Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Variables	No	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is migration flows between origin and destination provinces, estimated in levels using the Poisson pseudo–maximum likelihood (PPML) estimator. The sample restricts origin provinces to the 17 cities identified as most affected by drought conditions, while all provinces remain potential destinations. Temperature variables are measured in Kelvin (K) and precipitation in millimeters (mm). Distance is measured in kilometers (km), wildfire area in hectares (ha), electricity consumption in kilowatt-hours (kWh), and the refugee variable represents the percentage share of refugees in the origin province. All specifications include fixed effects for origin, destination, and year. Standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Destination climate conditions also influence migration decisions. The destination temperature enters the model with a positive, statistically significant coefficient, while the squared temperature term is negative and statistically significant. This pattern indicates a nonlinear temperature effect at the destination level, suggesting that migrants prefer moderately warm destinations but avoid areas experiencing extreme heat.

Economic variables further shape migration patterns. Origin GDP per capita is positive and statistically significant, suggesting that higher income levels may facilitate migration by easing the financial constraints households face. Destination GDP per capita also exhibits a positive, statistically significant coefficient, indicating that migrants are attracted to stronger economies.

Consistent with the gravity model framework, geographic distance exerts a negative and statistically significant effect in Models (5) and (6). The estimated coefficient of approximately −0.983 indicates that migration flows decline with increasing spatial distance.

Finally, the refugee share becomes positive and statistically significant in the full specification, with a coefficient of 0.052 ($p < 0.01$). This result suggests that demographic pressures may reinforce migration responses in drought-exposed provinces.

Overall, the explanatory power of the model increases as additional controls are introduced, with the pseudo-R^2 rising from 0.076 in Model (1) to 0.426 in Model (6). These results indicate that climatic conditions, particularly rising temperatures and declining precipitation, play an important role in shaping migration flows from drought-affected provinces.

Overall, the explanatory power of the model increases as additional controls are introduced, with the pseudo-R^2 rising from 0.076 in Model (1) to 0.426 in Model (6). The results confirm that climatic conditions, particularly temperature and precipitation shocks, remain important drivers of migration flows in provinces facing severe drought exposure.

7.4.3 Hausman–Taylor Estimation: 17-City Drought Subsample

Table 10 reports the Hausman–Taylor estimates for migration flows originating from the 17 provinces most affected by drought conditions in Türkiye over the period 2008–2024. The Hausman–Taylor estimator allows the inclusion of time-invariant variables such as geographic distance while addressing potential endogeneity of economic variables. In this specification, origin GDP per capita is treated as endogenous, whereas climate variables and refugee shares are considered exogenous shocks.

The results confirm the presence of nonlinear temperature effects within the drought-exposed provinces. Across specifications, the squared temperature term remains positive and statistically significant. In the full specification (Model 6), the coefficient on the squared temperature term equals 129.5 ($p < 0.01$), indicating that migration responses increase as temperatures move further away from moderate climatic conditions. This finding suggests that extreme heat plays an important role in generating migration pressures in drought-prone regions.

In contrast, precipitation variables are not statistically significant once other controls are introduced. Although the estimated coefficients are generally negative, their lack of statistical significance indicates that temperature-related stress and environmental shocks may play a more dominant role in shaping migration decisions in this highly exposed subsample.

Environmental shocks also contribute to migration dynamics. Beginning in Model (4), wildfire activity exhibits a positive and statistically significant coefficient. In the full specification, the estimated coefficient equals 0.00750 ($p < 0.01$), indicating that higher wildfire exposure is associated with increased migration outflows from drought-affected provinces.

Social pressures also appear to influence migration patterns in these regions. The refugee share variable becomes positive and statistically significant once introduced in Model (5). In the full specification, the coefficient equals 0.119 ($p < 0.01$), suggesting that higher refugee presence is associated with increased internal migration flows from these provinces.

Economic and geographic variables behave as expected. Origin GDP per capita, treated as endogenous, remains negative and statistically significant across specifications. In Model (6), the coefficient is −0.179 ($p < 0.01$), indicating that weaker economic conditions in origin provinces are associated with stronger outflows of migration. Distance also remains negative and statistically significant across all models, with an estimated coefficient of approximately −0.691, confirming the role of spatial frictions in internal migration.

Overall, the Hausman-Taylor estimates reinforce the main findings obtained from the fixed effects and PPML estimations. Nonlinear temperature effects, environmental shocks, and demographic pressures jointly contribute to migration dynamics in Türkiye's most drought-exposed provinces.

Table 10: Hausman-Taylor Estimates: Migration from 17 Drought-Exposed Provinces in Türkiye

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Origin Temperature (K)	9.373***	9.884***	9.307***	13.66***	9.919***	8.621***
	(1.441)	(1.442)	(1.450)	(2.169)	(2.232)	(2.262)
Origin Temperature^2		106.9***	101.4***	136.2***	134.7***	129.5***
		(44.60)	(44.68)	(62.33)	(62.77)	(63.07)
Origin Precipitation (mm)			-0.320	-0.165	-0.197	-0.126
			(0.217)	(0.231)	(0.231)	(0.233)
Origin Precipitation^2			0.0210	0.0101	0.0125	0.0076
			(0.0163)	(0.0177)	(0.0177)	(0.0179)
Wildfire Area (ha)				0.00829***	0.00740***	0.00750***
				(0.00229)	(0.00227)	(0.00227)
Origin GDP per capita (USD)	-0.342***	-0.341***	-0.342***	-0.176***	-0.106***	-0.179***
	(0.0312)	(0.0313)	(0.0313)	(0.0360)	(0.0372)	(0.0381)
Refugee Share (%)					0.121***	0.119***
					(0.0212)	(0.0213)
Electricity Consumption (kWh)						0.0575***
						(0.0171)
Distance (km)	-0.669***	-0.670***	-0.669***	-0.697***	-0.692***	-0.691***
	(0.0558)	(0.0558)	(0.0558)	(0.0543)	(0.0543)	(0.0544)
Constant	11.40***	11.10***	12.04***	10.97***	10.09***	9.652***
	(0.424)	(0.425)	(0.590)	(0.625)	(0.666)	(0.671)
Observations	23,120	23,120	23,120	16,320	16,320	16,320
Pair Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is the logarithm of internal migration flows between provinces. The sample restricts origin provinces to the 17 cities identified as most affected by drought conditions, while all provinces remain potential destinations. Estimations are conducted using the Hausman–Taylor (1981) instrumental variables estimator. Temperature variables are measured in Kelvin (K) and precipitation in millimeters (mm). Distance is measured in kilometers (km), wildfire area in hectares (ha), electricity consumption in kilowatt-hours (kWh), and the refugee variable represents the percentage share of refugees in the origin province. Origin GDP per capita is treated as endogenous, while climate variables and refugee shares are treated as exogenous. The time-invariant distance variable is identified through the internal instrumentation procedure of the Hausman–Taylor estimator. All specifications include origin–destination pair fixed effects and year fixed effects. Standard errors are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Overall, the empirical results across the different estimation approaches—three-way fixed effects, PPML, and Hausman–Taylor estimations—consistently indicate that climatic conditions are closely associated with internal migration patterns in Türkiye. The full-sample analysis reveals a nonlinear relationship between temperature and migration, suggesting that extreme heat in origin provinces increases migration pressures. The subsample analyses further show that these climate–migration relationships become more pronounced in provinces that are more exposed to environmental stress, particularly in drought-prone regions. In addition, environmental shocks such as wildfires and demographic factors such as refugee inflows appear to influence migration dynamics in specific regions. Taken together, these findings provide consistent empirical evidence that climatic variability, environmental shocks, and regional economic conditions jointly shape internal migration flows in Türkiye. The broader implications of these results are discussed in the following section.

8 Discussion

This study examines how climatic conditions, economic factors, and demographic pressures shape internal migration across Türkiye. Building on the Random Utility Model (RUM) and the gravity framework, the empirical analysis models migration flows as a function of economic size, climatic conditions, and spatial frictions between regions. The gravity model is particularly suitable for this analysis because it captures the proportional relationship between migration flows, the economic size of locations, and the distance between them. By incorporating origin and destination characteristics such as GDP, temperature, and precipitation, the framework allows us to examine how both economic opportunities and environmental conditions influence migration decisions.

To estimate these relationships, we employ three complementary econometric approaches: a three-way fixed effects specification, the Poisson Pseudo-Maximum Likelihood (PPML) estimator, and the Hausman–Taylor estimator. The three-way fixed effects model controls for unobserved heterogeneity across origins, destinations, and time. The PPML estimator provides robustness to zero migration flows and heteroskedasticity, yielding consistent estimates even when many city pairs report no migration (Silva and Tenreyro, 2006a). Finally, the Hausman–Taylor estimator addresses potential endogeneity and allows the estimation of both time-varying and time-invariant variables, such as geographic distance, by decomposing the error term into individual-specific and idiosyncratic components (Hausman and Taylor, 1981; Wooldridge, 2010). Across these specifications, the results consistently highlight the importance of climatic stress—particularly rising temperatures and declining precipitation—in shaping internal migration dynamics.

A central finding of the analysis is the presence of nonlinear temperature effects. Across several specifications, especially in the full sample and the drought-exposed subsamples, the squared temperature term remains positive and statistically significant. This pattern suggests that migration responses intensify as temperatures deviate further from moderate climatic conditions. In practical terms, moderate increases in temperature may have limited effects on migration decisions, but extreme heat substantially increases migration pressures. These findings align with the "voting with their feet" mechanism, where individuals relocate to escape deteriorating environmental conditions (Rappaport, 2009; Backhaus et al., 2015; Cai et al., 2016; Thiede et al., 2016). In Türkiye, where southern regions already experience high summer temperatures, further warming appears to be a strong push factor for migration toward more temperate regions.

An additional insight from the graphical and bin-based analysis concerns the presence of a clear thermal threshold in migration responses. Using temperature bin specifications and seasonal measures of climate exposure, we find that migration probabilities increase sharply once summer maximum temperatures exceed approximately 32°C. Below this level, migration responses are relatively moderate, suggesting that households can adapt to typical climatic fluctuations. However, when temperatures move beyond this threshold, migration becomes substantially more likely. This pattern is consistent with the nonlinear temperature effects identified in the regression models and indicates the presence of a "thermal tipping point" in internal migration decisions.

The seasonal analysis further reveals that migrants respond differently to summer and winter temperature conditions. High summer temperatures in origin regions appear to act as strong push factors, while relatively milder winter temperatures at destination locations increase their attractiveness. These findings support the interpretation that migration decisions are shaped not only by average climatic conditions but also by seasonal extremes. The existence of a temperature threshold around 32°C suggests that future warming may generate disproportionately large migration responses once local climates cross critical thermal limits.

The role of precipitation in shaping migration flows is also notable. In contrast to earlier expectations that higher precipitation might facilitate migration through improved agricultural productivity, our empirical results suggest that lower rainfall levels are more consistently associated with increased out-migration. In several specifications, particularly within the drought-exposed subsample, precipitation exhibits a negative and statistically significant relationship with migration flows. These findings are consistent with the theoretical predictions embedded in our econometric framework, where reduced rainfall contributes to environmental stress, agricultural losses, and declining local economic conditions. Such pressures increase incentives for households to migrate in search of more stable living conditions. These results are consistent with previous studies linking drought conditions and environmental degradation to migration decisions (Backhaus et al., 2015; Hirvonen, 2016; Beine and Parsons, 2017; Mahajan and Yang, 2020).

Environmental shocks also contribute to migration dynamics. In several specifications, wildfire exposure is positively and statistically associated with migration flows, suggesting that regions experiencing larger wildfire events tend to generate higher out-migration. Wildfires represent sudden environmental disruptions that can damage housing, agricultural land, and local infrastructure, thereby reducing the livability of affected areas and increasing incentives to relocate. These findings are consistent with broader evidence that environmental disasters can accelerate migration by disrupting local economic activity and increasing uncertainty about future living conditions.

Electricity consumption also emerges as an important variable in several specifications. The positive association between electricity consumption and migration can reflect multiple mechanisms. On the one hand, higher electricity usage may serve as a proxy for regional economic development and income levels, thereby relaxing financial constraints and enabling households to migrate. On the other hand, electricity consumption may capture local adaptation capacity, as increased energy use often reflects greater reliance on cooling technologies, irrigation systems, and other infrastructure designed to mitigate climate stress. In this sense, electricity consumption may simultaneously reflect both economic opportunity and adaptive responses to rising temperatures. The positive relationship observed in the data suggests that regions with greater economic capacity or infrastructure may facilitate migration by lowering the costs of relocation and enabling households to respond more flexibly to climatic pressures.

Our subsample analyses further reveal that climate-induced migration pressures are not uniform across regions. Provinces identified as highly exposed to climatic stress—particularly those located in Türkiye's southeastern and Mediterranean regions—exhibit stronger migration responses to temperature increases and precipitation declines. The results for the drought-exposed subsample of 17 provinces indicate that both environmental shocks and demographic pressures interact to amplify migration dynamics in these vulnerable areas. These findings suggest that climatic shocks have the largest migration impacts in regions already facing environmental and socioeconomic constraints.

Climate change further exacerbates environmental stressors such as droughts, which contribute to resource scarcity and competition. These pressures can heighten the risk of conflict, as illustrated in Syria, where prolonged droughts weakened agricultural systems and contributed to civil unrest (Kelley et al., 2015). Such conflicts force people to migrate in search of safety, creating large-scale displacement. The resulting refugee influx has strained host communities and generated secondary migration, as both refugees and locals relocate in search of opportunities (Rapoport et al., 2020). Climatic shocks that reduce agricultural yields can drive food insecurity, increasing the likelihood of violence (Raleigh, 2011; Bozkurt and Sen, 2013; Moore and Wesselbaum, 2023). For instance, Hendrix and Salehyan (2012) demonstrates that both extreme wet and dry conditions in Africa are correlated with civil strife, further linking climate variability to conflict.

The presence of Syrian refugees also plays an important role in shaping migration patterns within Türkiye. Since the outbreak of the Syrian civil war in 2011, Türkiye has hosted one of the largest refugee populations in the world. The arrival of millions of refugees has placed pressure on housing markets, labor markets, and local public services in several provinces. Our results show that higher refugee shares are associated with increased internal migration flows in certain specifications, suggesting that demographic pressures may reinforce migration decisions among the domestic population. These patterns highlight the importance of considering both environmental and demographic factors when analyzing migration dynamics (Sagiroglu; Erdoğan and Cantürk, 2022).

The interaction between climatic shocks and conflict-driven displacement provides an important contextual backdrop for these findings. Previous research has highlighted how prolonged drought conditions contributed to agricultural collapse and social instability in Syria prior to the outbreak of civil war (Kelley et al., 2015). These environmental pressures triggered large-scale rural displacement and exacerbated underlying economic vulnerabilities. As conflict intensified, millions of Syrians were forced to flee the country, with a substantial share settling in Türkiye. According to UNHCR,[22] more than 14 million Syrians have been displaced since 2011, with approximately four million residing in Türkiye. The resulting refugee inflows have reshaped demographic patterns and contributed to secondary migration dynamics within host communities (Rapoport et al., 2020).

Environmental stress and demographic pressures can also contribute to broader social tensions and economic adjustments. Large population inflows may strain local infrastructure, public services, and labor markets, while migrants themselves face challenges related to social integration and economic adaptation. Previous studies emphasize that migration responses to environmental shocks often interact with social and political dynamics, reinforcing the complexity of climate-related migration processes (Koubi et al., 2016; Rapoport et al., 2020; Adida et al., 2017; Gillis et al., 2023). These interactions highlight the importance of understanding migration not solely as a response to environmental change but as part of a broader socioe-

[22]Data source: https://www.unrefugees.org/

conomic adjustment process.

Overall, the findings of this study emphasize the multifaceted relationship between climate variability, economic conditions, and migration dynamics. Rising temperatures, declining precipitation, and environmental shocks appear to increase migration pressures, particularly in regions already facing environmental vulnerability. At the same time, demographic pressures such as refugee inflows may further reshape migration patterns by altering local economic and social conditions. These results underscore the importance of integrated policy approaches that address both environmental risks and demographic pressures when designing strategies to manage internal migration in the context of climate change.

9 Conclusion

This study examines the relationship between climatic conditions and internal migration in Türkiye using a gravity-based framework grounded in the Random Utility Model. Using province-level panel data for the period 2008–2024, we estimate migration responses with three complementary econometric approaches: a three-way fixed effects specification, the Poisson Pseudo-Maximum Likelihood (PPML) estimator, and the Hausman–Taylor estimator. These approaches allow us to account for spatial heterogeneity, zero migration flows, and potential endogeneity in the estimation of migration dynamics.

The empirical results provide consistent evidence that climatic stress plays an important role in shaping internal migration patterns in Türkiye. In particular, we find strong nonlinear temperature effects, indicating that migration responses intensify as temperatures deviate from moderate climatic conditions. The graphical and bin-based analyses further reveal the presence of a thermal threshold: migration flows increase sharply once summer maximum temperatures exceed approximately 32°C. These results suggest that extreme heat acts as a powerful push factor for migration, particularly in regions already exposed to high climatic stress.

Precipitation also plays an important role in migration decisions. Across several specifications, lower rainfall levels are associated with higher migration outflows, particularly in provinces facing severe drought exposure. These findings suggest that declining precipitation and water scarcity can increase migration pressures by weakening agricultural productivity and local economic stability.

The results further highlight the importance of environmental shocks and socioeconomic factors in shaping migration dynamics. Wildfire exposure is positively associated with migration flows, indicating that sudden environmental disruptions can accelerate population movements. Electricity consumption also exhibits a positive relationship with migration, which may reflect both higher income levels and greater adaptation capacity through infrastructure such as cooling technologies and irrigation systems. In addition, refugee presence in origin regions appears to influence internal migration in some specifications, suggesting that demographic pressures may interact with environmental stress to shape mobility patterns.

In addition to climatic stressors, the analysis highlights the role of demographic pressures associated with refugee inflows. Although refugee movements are not the primary focus of this study, the results suggest that refugee presence can operate as a secondary exposure that reshapes local migration dynamics. In several specifications, higher refugee shares are associated with increased internal migration flows, particularly in environmentally vulnerable provinces. This pattern is consistent with the idea of secondary displacement, where environmental stress and demographic pressures jointly influence mobility decisions. In the context of Türkiye, the large inflow of Syrian refugees following the civil war has altered regional population dynamics and may have intensified migration pressures in provinces already facing climatic challenges. These findings suggest that climate-related migration cannot be fully understood without considering the broader demographic and institutional context in which environmental shocks occur.

Subsample analyses reinforce these findings by showing that climate-induced migration pressures are particularly strong in provinces highly exposed to climatic stress and drought. These regions exhibit stronger migration responses to rising temperatures and declining precipitation, highlighting the importance of regional vulnerability in shaping climate-related migration outcomes.

Overall, the findings underscore the complex interaction between climate variability, environmental shocks, and socioeconomic factors in driving internal migration in Türkiye. As climate change continues to intensify temperature extremes and water scarcity in many parts of the country, migration pressures may become increasingly concentrated in environmentally vulnerable regions. Future research could build on these findings by incorporating higher-frequency climatic data, exploring household-level migration responses, and examining how adaptation policies influence mobility decisions in climate-exposed regions.

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A Additional Figures

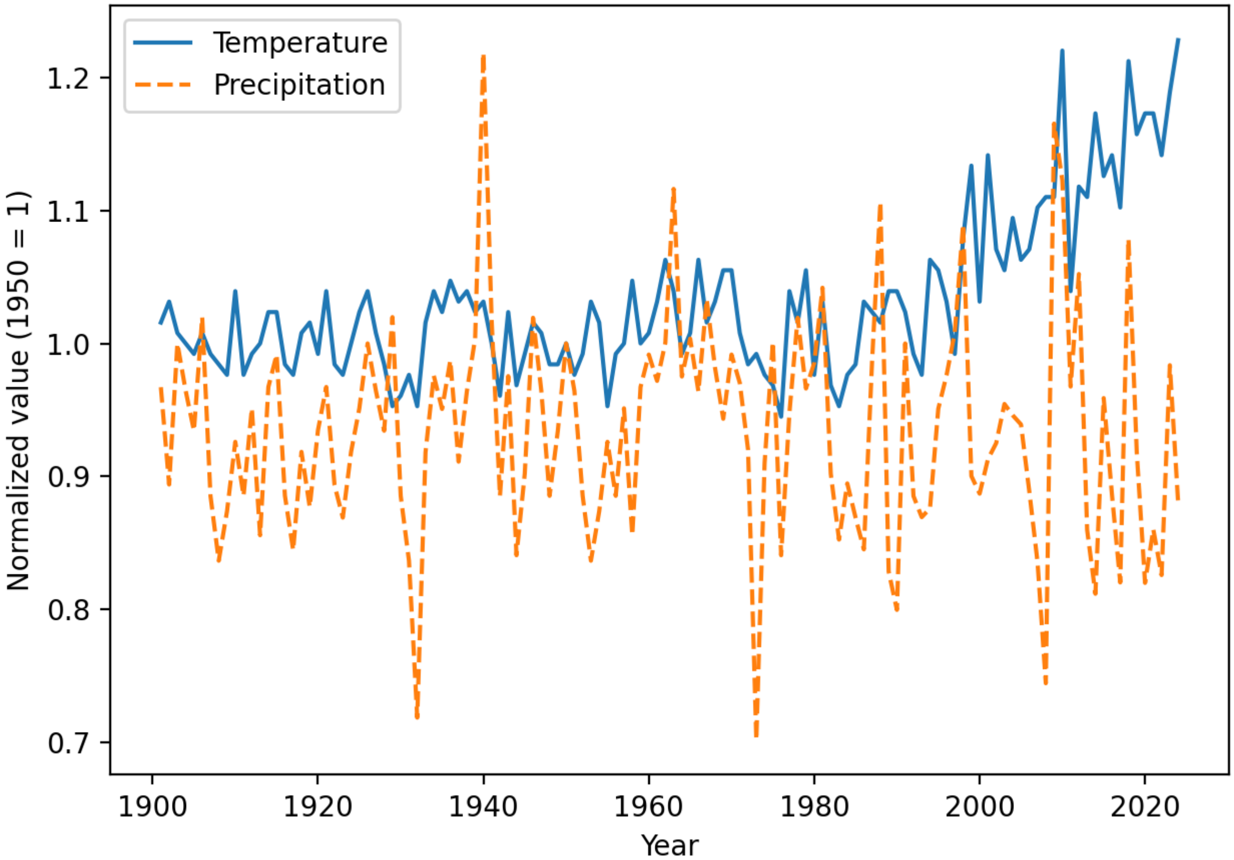

Figure A1: Long-run trends in average temperature and precipitation in Türkiye over the period 1901–2024. Both series are normalized to their 1950 values to facilitate comparison across variables measured in different units.

Figure A2: Trends in average temperature and precipitation in Türkiye for the period 1970–2024, normalized to 1950. The figure highlights the acceleration of temperature increases in the post-1970 period.

B Supplementary Results Figures

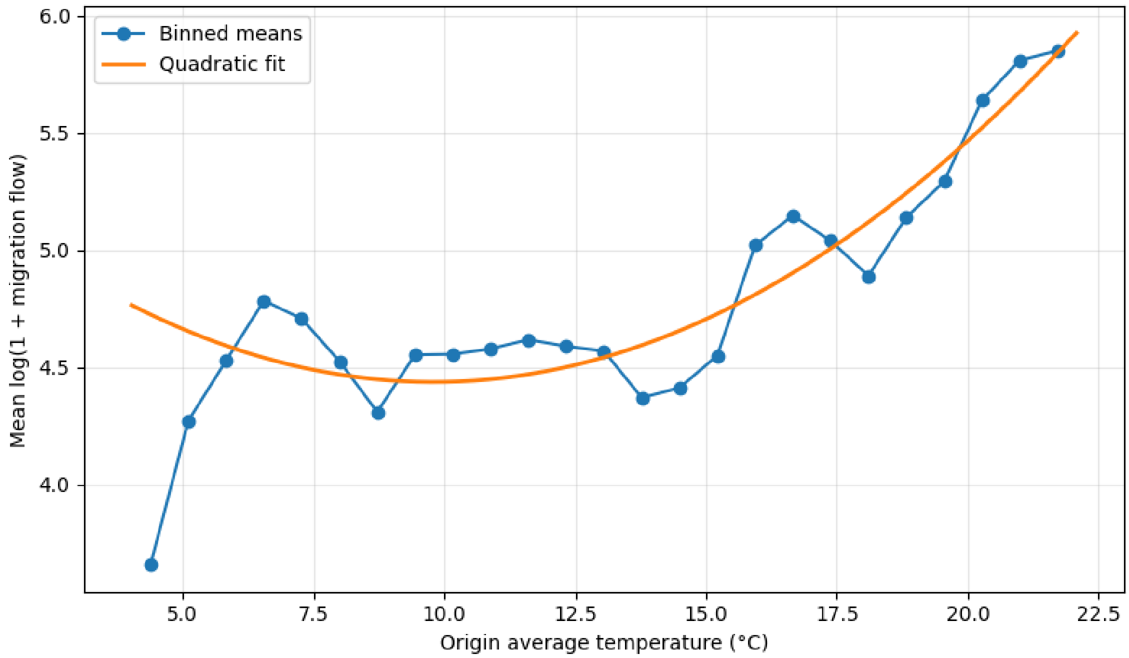

Figure B1: Migration Flow and Origin Average Temperature

Note: This figure plots binned averages of the logarithm of one plus migration flows against origin average temperature. The solid line represents a quadratic fit.

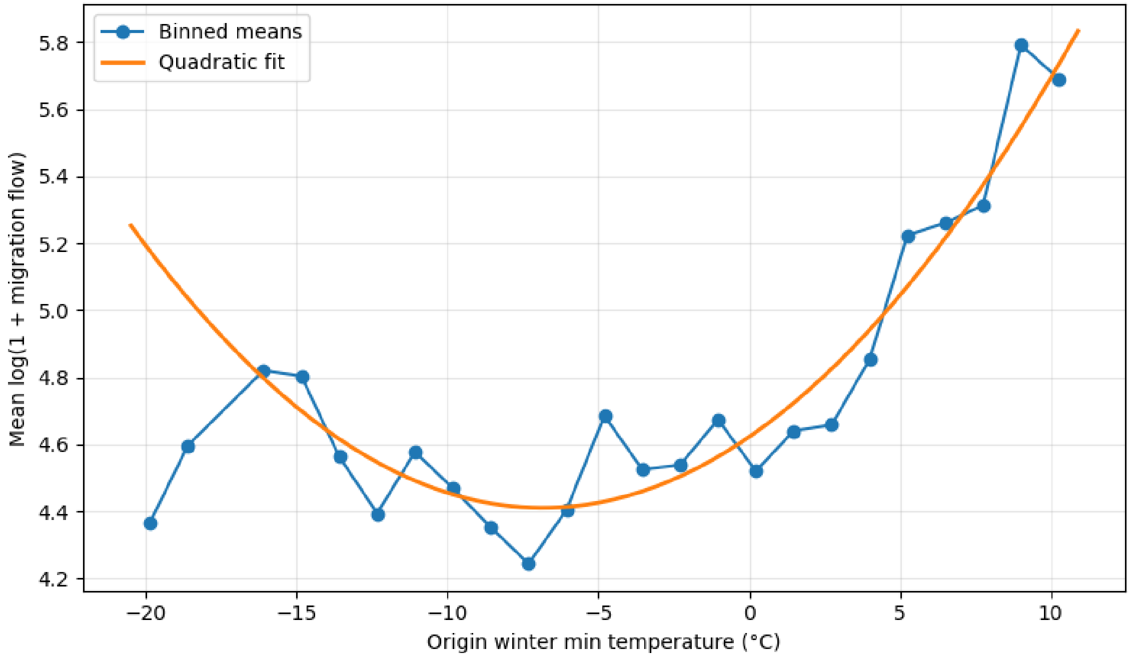

Figure B2: Migration Flow and Origin Winter Minimum Temperature

Note: This figure plots binned averages of the logarithm of one plus migration flows against origin winter minimum temperature. The solid line represents a quadratic fit.

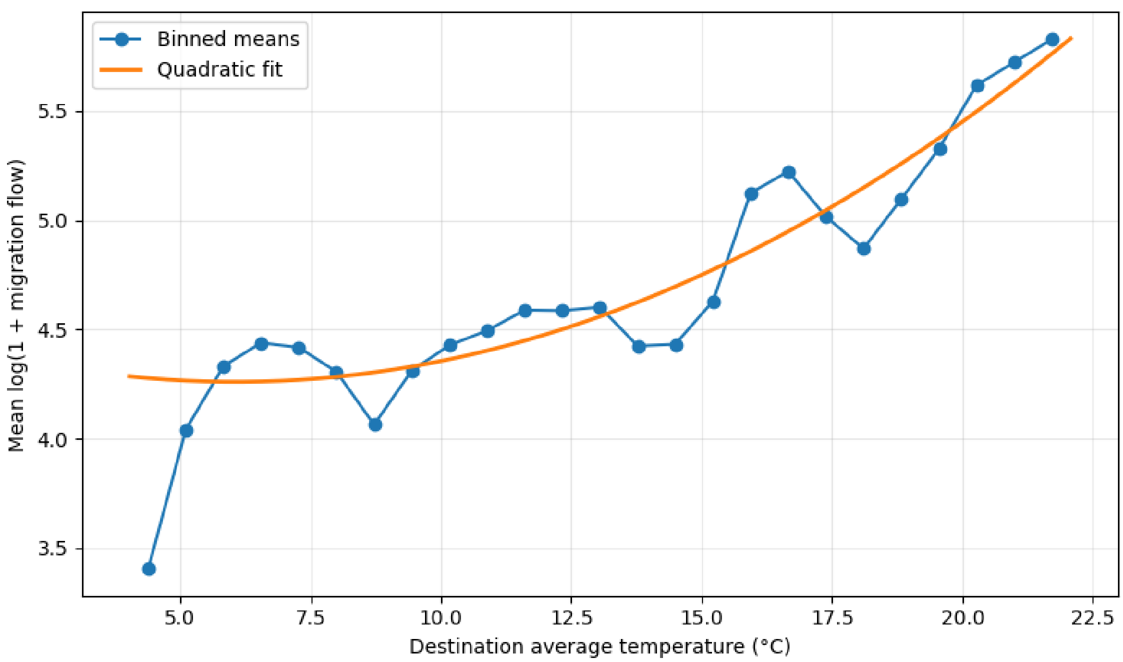

Figure B3: Migration Flow and Destination Average Temperature

Note: This figure plots binned averages of the logarithm of one plus migration flows against destination average temperature. The solid line represents a quadratic fit.

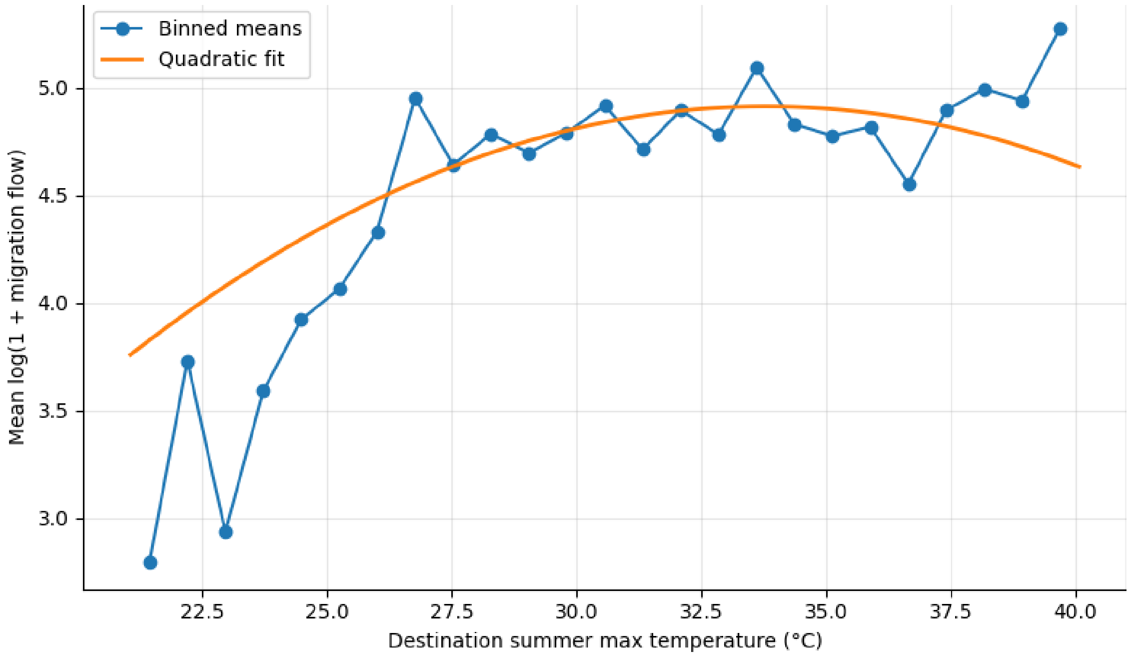

Figure B4: Migration Flow and Destination Summer Maximum Temperature

Note: This figure plots binned averages of the logarithm of one plus migration flows against destination summer maximum temperature. The solid line represents a quadratic fit.

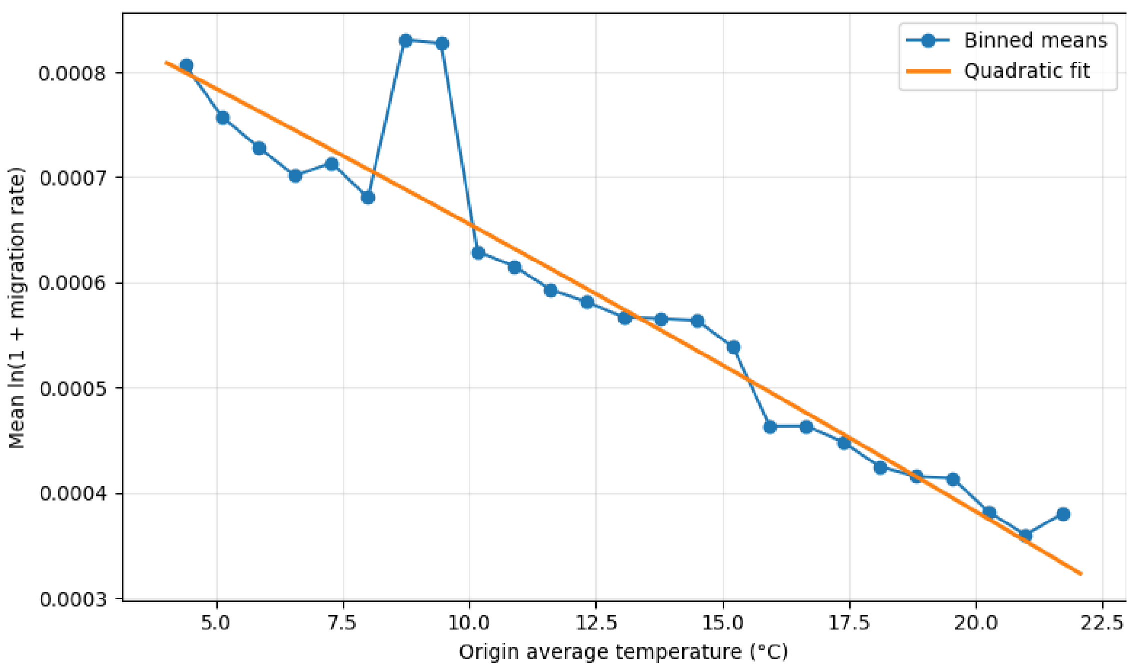

Figure B5: Migration Rate and Origin Average Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against origin average temperature. Migration rates are defined as migration flows divided by origin population. The solid line represents a quadratic fit.

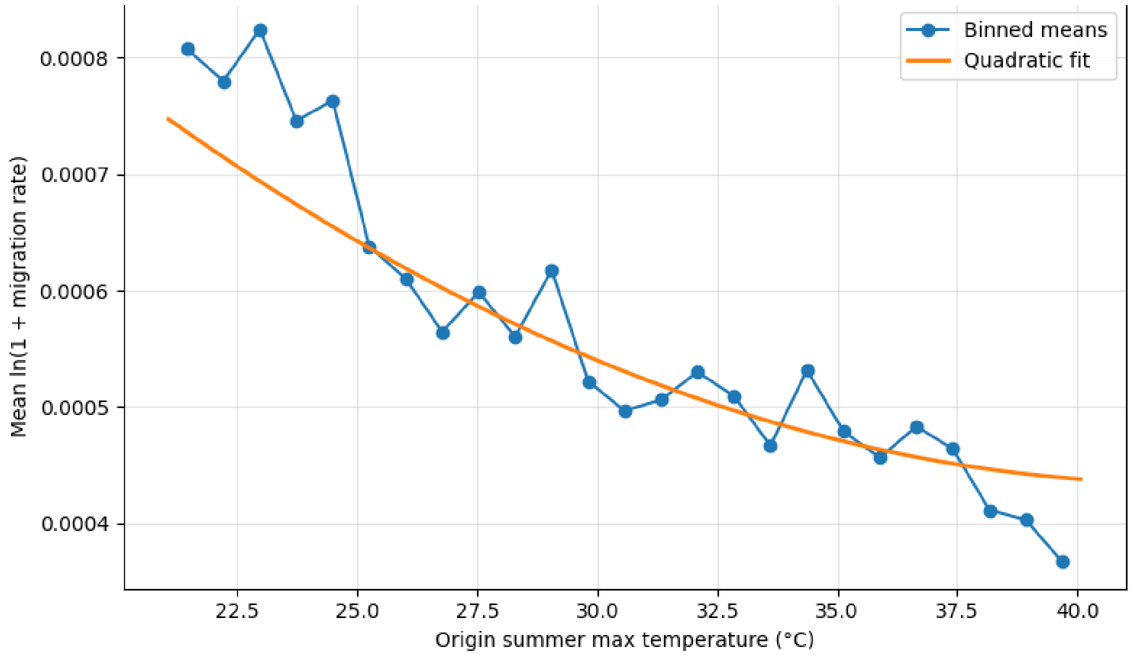

Figure B6: Migration Rate and Origin Summer Maximum Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against origin summer maximum temperature. Migration rates are defined as migration flows divided by origin population. The solid line represents a quadratic fit.

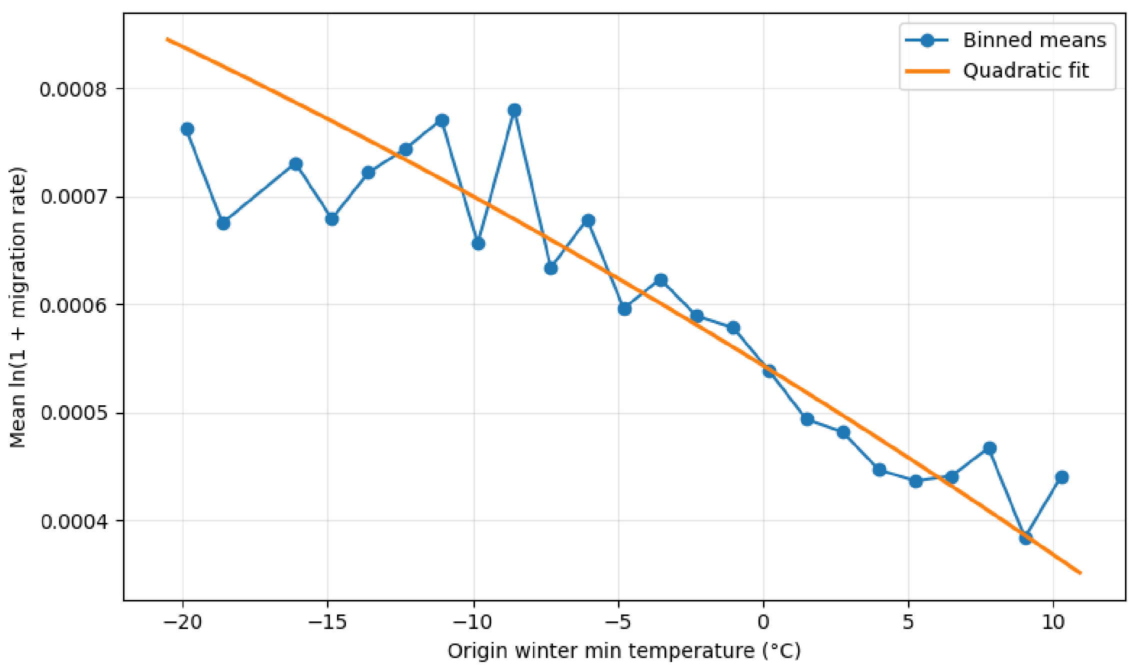

Figure B7: Migration Rate and Origin Winter Minimum Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against origin winter minimum temperature. Migration rates are defined as migration flows divided by origin population. The solid line represents a quadratic fit.

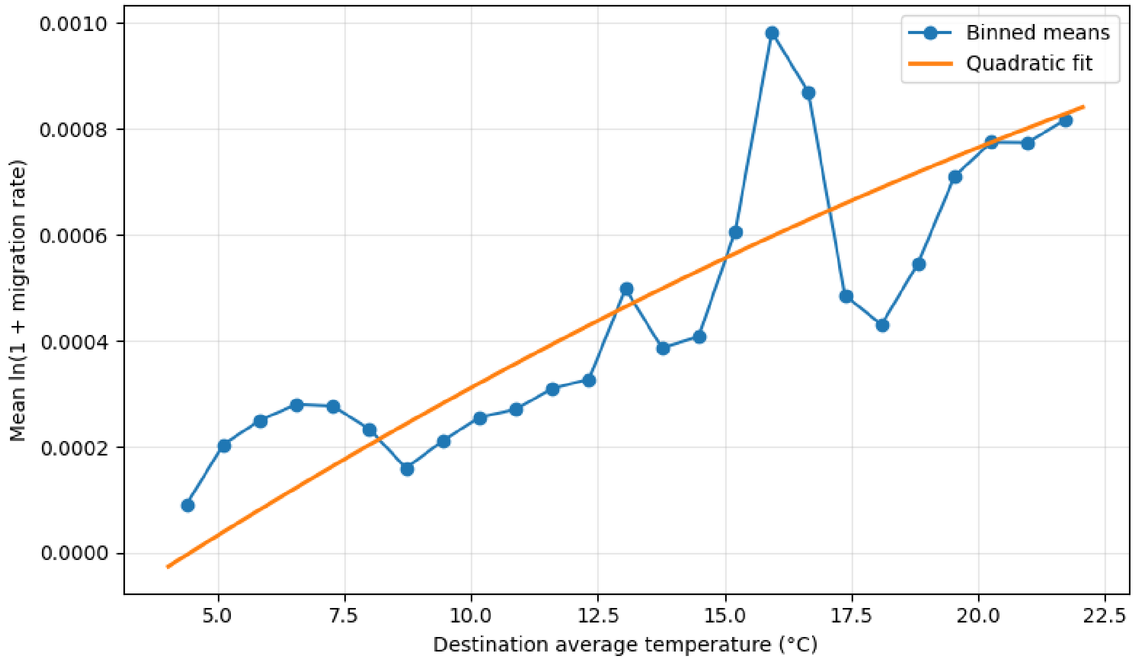

Figure B8: Migration Rate and Destination Average Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against destination average temperature. Migration rates are defined as migration flows divided by origin population. The solid line represents a quadratic fit.

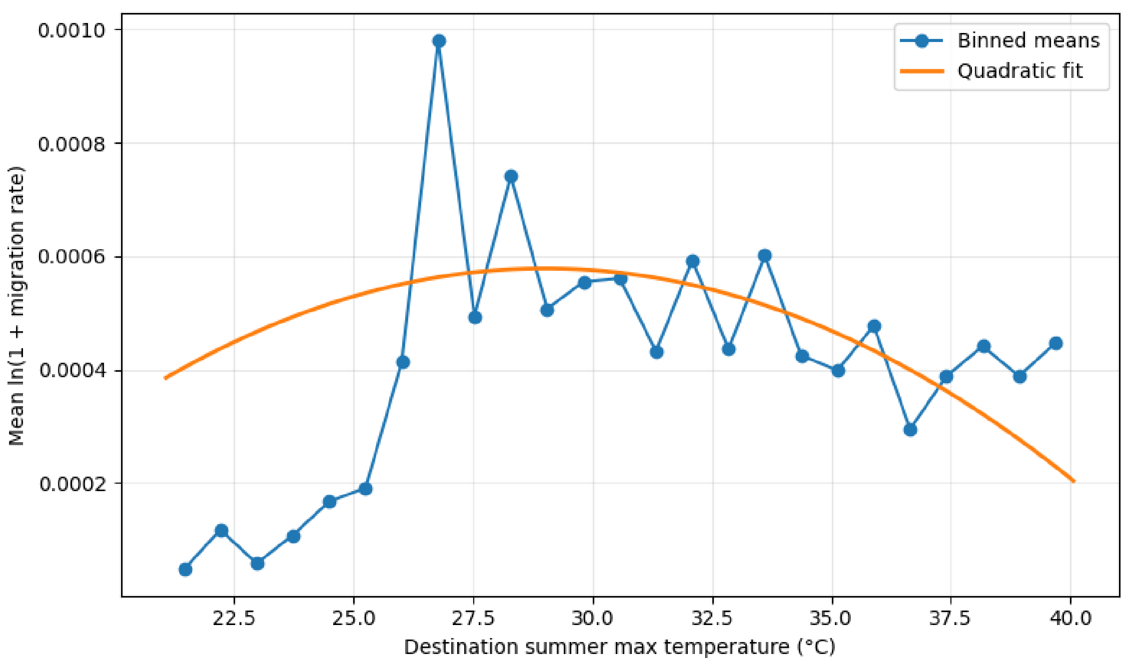

Figure B9: Migration Rate and Destination Summer Maximum Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against destination summer maximum temperature. Migration rates are defined as migration flows divided by origin population. The solid line represents a quadratic fit.

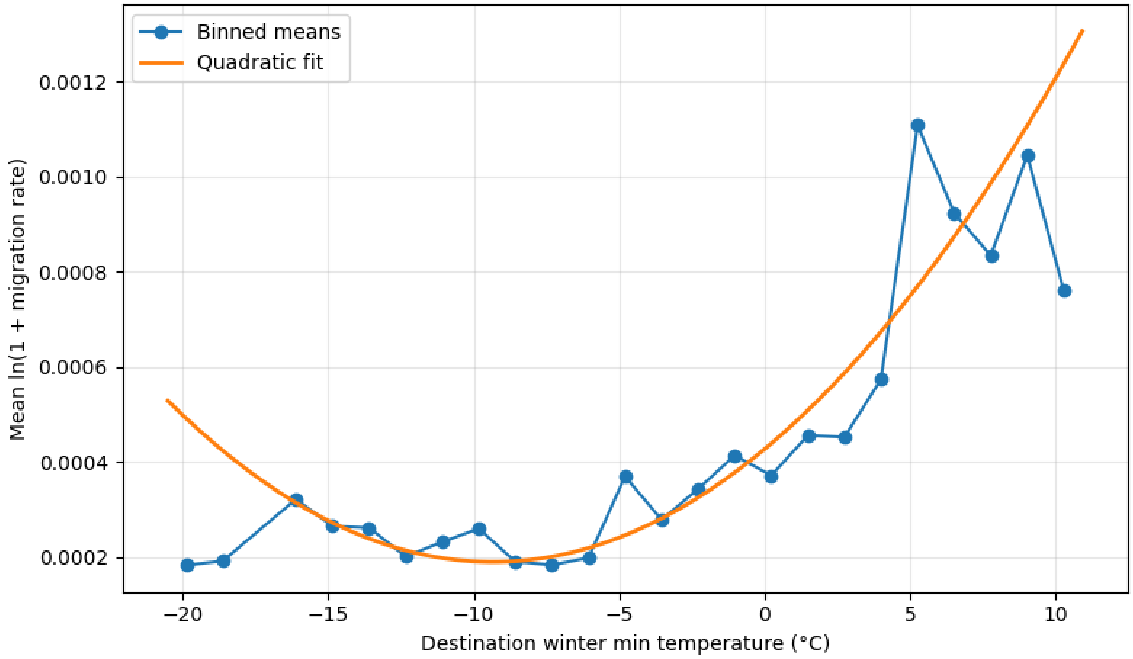

Figure B10: Migration Rate and Destination Winter Minimum Temperature

Note: This figure plots binned averages of the logarithm of one plus the migration rate against destination winter minimum temperature. Migration rates are defined as migration flows divided by origin population. The solid line represents a quadratic fit.

C Temperature Bin Analysis

To examine whether migration responses change once temperature thresholds are crossed, we estimate a temperature-bin specification using pooled OLS with origin, destination, and year fixed effects. Instead of modeling temperature using quadratic terms, temperatures are grouped into bins representing extreme cold and extreme heat conditions. Moderate temperature ranges serve as the reference category.

This specification allows us to evaluate whether migration flows respond differently when provinces experience extreme climatic conditions. Temperatures below 0°C represent extremely cold winter conditions, while temperatures above 33°C represent extreme summer heat.

Table C1 presents the results for the full sample of 81 provinces over the period 2008–2024. The estimates show that high temperatures in origin provinces increase migration flows, suggesting that extreme heat acts as a push factor. In contrast, extremely high temperatures in destination provinces reduce migration flows, indicating that migrants tend to avoid excessively hot destinations.

Table C1: Temperature Bin Results for Internal Migration: 81 Provinces in Türkiye (2008–2024)

	Model 1	Model 2	Model 3	Model 4
Origin Temperature				
Average Temperature (K)	4.309***	3.983***	4.128***	–
	(0.817)	(0.842)	(0.838)	
Cold Bin (<0°C)	–	–	–	-0.019***
				(0.005)
Hot Bin (>33°C)	–	–	–	0.014**
				(0.006)
Destination Temperature				
Average Temperature (K)	–	–	11.660***	–
			(0.891)	
Cold Bin (<0°C)	–	–	–	-0.005
				(0.004)
Hot Bin (>33°C)	–	–	–	-0.012**
				(0.006)
Control Variables				
Distance (km)	-0.787***	-0.787***	-0.787***	-0.787***
	(0.014)	(0.014)	(0.014)	(0.014)
Origin Precipitation (mm)	–	-0.015***	-0.015***	-0.020***
		(0.005)	(0.005)	(0.005)
Destination Precipitation (mm)	–	–	0.013***	0.002
			(0.005)	(0.005)
Origin Fixed Effects	Yes	Yes	Yes	Yes
Destination Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Observations	110,160	110,160	110,160	110,160
Moderate Temperature Bin (Reference)	–	–	–	Yes

Note: The dependent variable is the logarithm of internal migration flows between provincial pairs. Estimations are conducted using pooled OLS with origin, destination, and year fixed effects. Temperature bins are defined in degrees Celsius (°C), where cold bins correspond to temperatures below 0°C and hot bins correspond to temperatures above 33°C; moderate temperatures serve as the reference category. Origin temperature variables capture potential climate push factors, while destination temperature variables capture potential pull effects. Standard errors clustered at the origin–destination pair level are reported in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.